MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project Mansourah Massarah Gold Project
Customer: MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: DESIGN CALCULATION
49 FE DISEL, PETROL & JET FUEL STORAGE AREA
DESIGN OF FOUNDATION

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
Typ.	:	Typical
HCS	:	Hollow Core Slab
Fc'	:	Concrete Compressive Strength
Fy	:	Yield Strength of Reinforcing Steel
ϕ	:	Strength Reduction Factor
SBC	:	Saudi Building Code
SASO	:	Saudi Standards, Metrology and Quality Organization
ACI	:	American Concrete Institute
ASCE	:	American Society of Civil Engineers
PCI	:	Precast/Prestressed Concrete Institute

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1. INTENT

This Document is intended to produce the technical needs in Engineering including particular reference to design justification & Engineering validation works. The work described below is related to Design of foundation of equipment and miscellaneous pipe supports in Fuel Oil Storage area.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The Scope of this document is limited to the foundation design of storage tank, pumps and miscellaneous pipe supports.

4. DESIGN METHODOLOGY & ASSUMPTIONS

Storage tank GA and saddle layout is taken from available mechanical drawing Foundation loadings are calculated manually. Foundation design is performed using excel sheet with calculated loads.

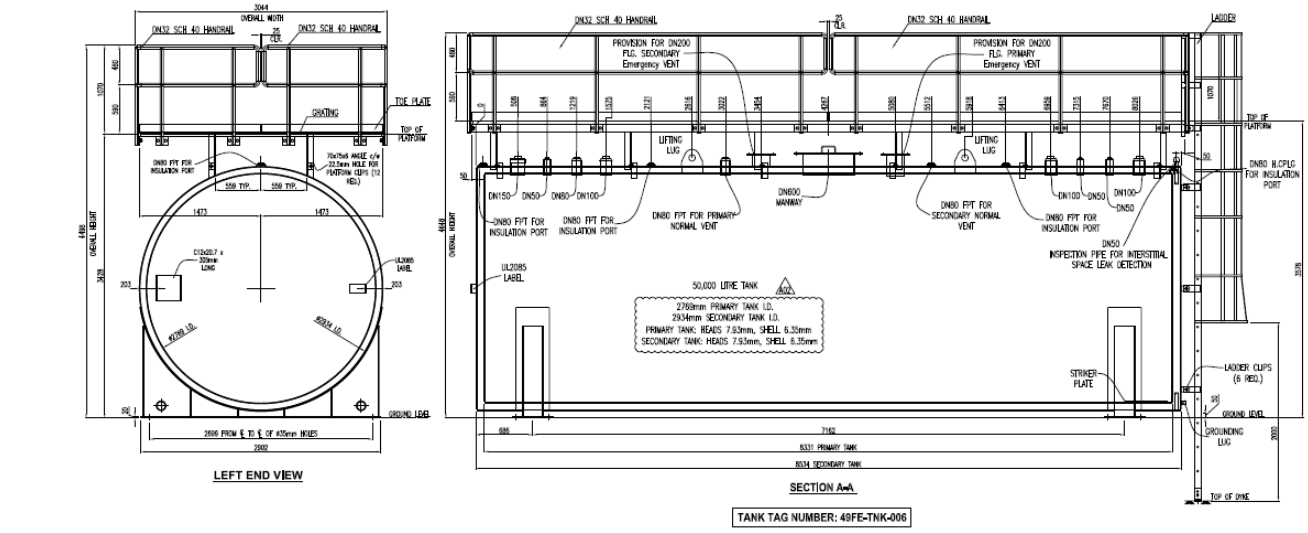
For miscellaneous pipe supports loadings are calculated based on input provided in overall layout drawing.

For pumps detail dynamic analysis not performed. Foundation size arrived based on base plate arrangement and foundation mass ratio requirement.

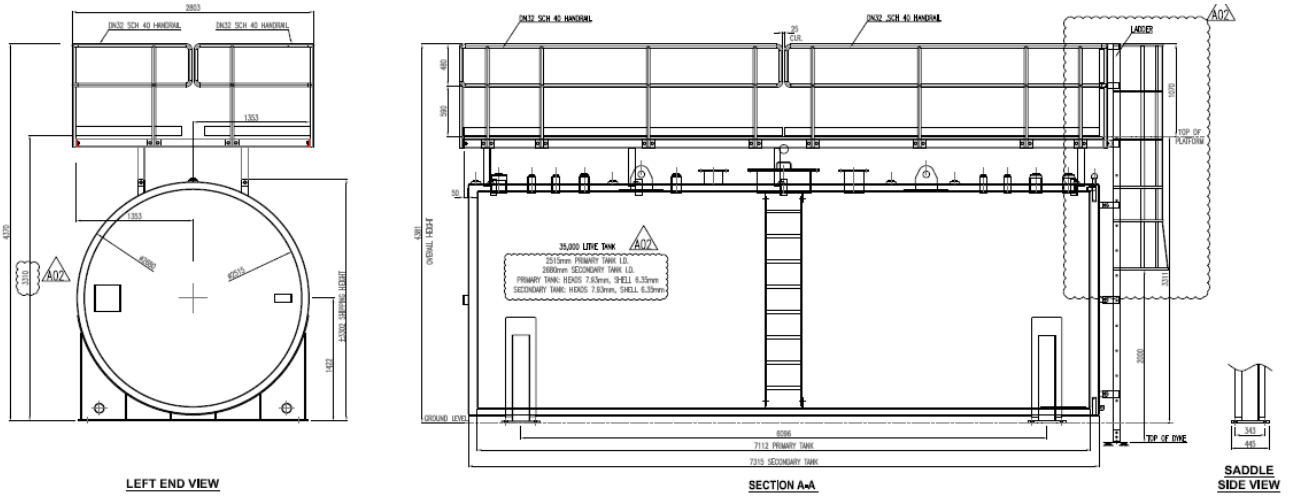
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49FE-TNK-006

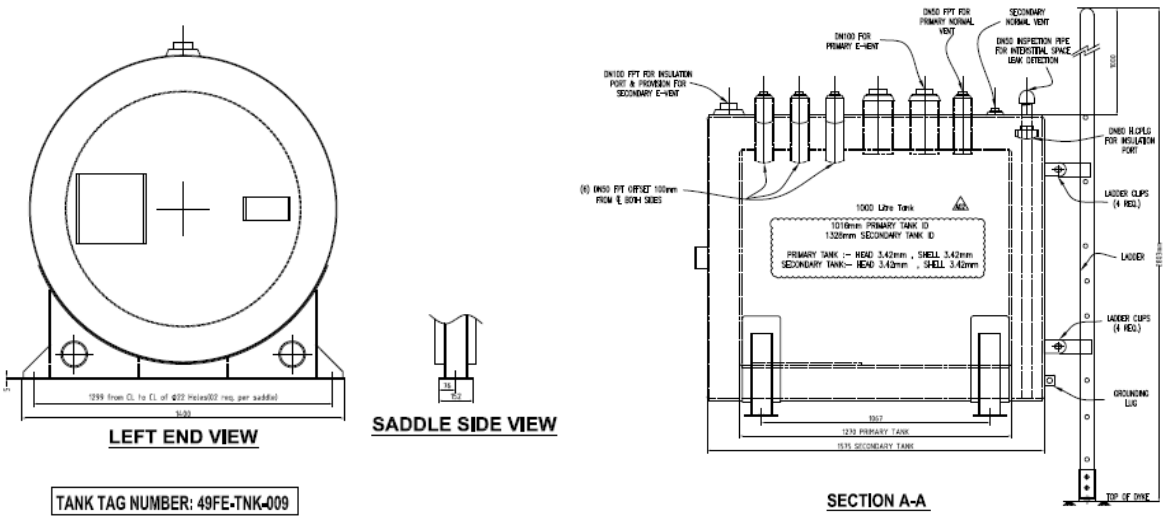


49FE-TNK-003

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49FE-TNK-009

4.1. Geotechnical investigations

Foundations have been designed based on recommendations of the Soil Investigation Report, Doc # MD-216-3000-EG-CV-RPT-1008.

4.2. Material Properties

Compressive Strength of Concrete at 28th days

fc' = 35 MPa (5.00 Ksi) for beams, columns, suspended slabs.

fc' = 35 MPa (5.00 Ksi) for foundations, grade beams, columns.

Yield Strength of Steel

fy = 414 MPa (60 Ksi) for all reinforcing steel.

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5. DESIGN LOADS

5.1. Dead Load(DL)

5.1.1. Empty, Operating & Test Load

	49FE-TNK-003	49FE-TNK-006	49FE-TNK-009
Empty Load, De (kN)	1.1 x 121.4 = 133.6	1.1 x 158.5 = 174.5	1.1 x 13.85 = 15.3
Operating Load, Do (kN)	1.1 x 481.95 = 530.15	1.1 x 659.75 = 725.75	1.1 x 23.95 = 26.5
Test Load, Dt (kN)	Considered Same as Operating Weight		

Equipment loads are obtained from vendor drawing. (See attachment 1).

Note: Tank weight supplied by manufacturer does not include the weight of attached pipe, platform etc. An increase of 10% weight is considered to account for above mentioned items.

5.1.2. Transverse Moment from eccentricity:

Eccentricity considered, e = (basic diameter/2 + 1.5ft) m (based on Process Industry practice guideline)

	Eccn, e (m)	Empty trans. MTe (kN.m)	Operating Trans. MTo (kN.m)	Test Trans. MTt (kN.m)
49FE-TNK-003	2.68/2+0.45 = 1.79	0.1x133.6x1.79 = 23.92	0.1x530.15x1.79 = 94.9	Same as operating
49FE-TNK-006	2.94/2+0.45 = 1.92	0.1x174.5x1.92 = 33.5	0.1x725.75x1.92 = 139.4	Same as operating
49FE-TNK-009	1.33/2+0.45 = 1.12	0.1x15.3x1.12 = 1.72	0.1x26.5x1.12 = 2.97	Same as operating

Note: Based on CG of the equipment shown in drawing (see attachment 1), dead load is distributed equally in both saddle.

5.1.3. Live Loads (LL):

Live loads are with reference to Clause-4.2.2, MD-216-1000-EG-ST-CRT-1001

Access way = **5 kN/m²**

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Area of platform attached = 9m x 3m. (49FE-TNK-003 & 49FE-TNK-006)

Hence, total LL = 5 x 9 x 3 = 135 kN.

Live load distributed equally in both saddles.

5.1.4. Thermal Force (Tf):

(a) Compute force required to deflect pier:

	Pier Size	I, mm^4	E (kN/mm^2)	Temp	ε, mm/mm	L, m	εL, mm	Force, T, kN
49FE-TNK-003	2.855x0.65x2.1	6.534 x 10^{10}	24.84	50 ^{0}C	0.00035	6.096	2.134	561
49FE-TNK-006	3.1x0.65x1.9	7.095 x 10^{10}			0.00035	7.612	2.66	1027
49FE-TNK-009	1.6x0.35x1.6	0.572 x 10^{10}			0.00035	1.067	0.373	20

Ambient temperature variation considered. Hence, maximum 50ºc temperature is considered for thermal load calculation.

Thermal expansion coefficient for carbon steel at 93.3^{0}C, ε = 0.00035 mm/mm (see attachment 2).

Thus, $T = 3(\varepsilon L)\ EI/2H^3 = (3 \times (\varepsilon L) \times 24.84 \times EI)/(2 \times 1700^3)$

(b) Compute sliding force assuming steel to steel friction:

	Operating Weight on one saddle	μ	Ff
49FE-TNK-003	266 kN	0.4	106.5 kN
49FE-TNK-006	363 kN	0.4	146 kN
49FE-TNK-009	13.5 kN	0.4	5.5 kN

Hence, Ff<T, on pedestal due to temperature

5.2. Wind Load (WL)

All structures and foundations (except for buried structures) designed and shall be constructed to resist the wind effects determined in accordance with Section-7.2, SBC-301-Analytical Procedure.

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The Wind Loads calculated here are in accordance with SBC-301-2007-Structural Loading & Forces.

General wind load requirements:

Basic wind speed:	V = 49 m/s (or) 177 kmph (Refer Clause-4.2.6, Doc-MD-216-1000-EG-ST-CRT-1001)
Risk category:	II (As per SBC-301, Table 1.6-1)
Velocity pressure coefficient:	K_d = **0.85** (As per SBC-301, Table 6.4-1)
Exposure category:	D (As per SBC-301, Section 6.4.2.3)
Enclosure classification:	Enclosed buildings
Gust effect factor:	G_f = **0.85** (As per SBC-301, Section-7.2.7.1)
Importance factor:	1 (As per SBC-301, Section-9.5)
Topography factor:	Kzt = 1.0 (As per SBC-301, Section 6.4.3.2)

Thus, $q_z = .0473 \times 10^{-3}$ Kz Kzt Kd V^2 I (Ref equation 7.2-12, SBC 301) .

$= 1.30$ kN/m^2

Wind induced forces = $q_z \times C_f \times G \times A_f$ kN

5.2.1. Wind Perpendicular to tank:

	Tank Dia, D mm	Height above Ground , h, m	h/D	D x sqrt(q_z)	Cf	Projected Area, A, m2	Wind Force, kN	Force on each pier, kN	Moment on Pedestal Top, kNm
49FE-TNK-003	2680	2.5	0.93	3.06	0.7	7.315x2.68 = 19.61	15.17	7.6	15.2
49FE-TNK-006	2934		0.85	3.35	0.7	8.534x2.93 4 = 25.04	19.5	9.75	19.5
49FE-TNK-009	1328		1.9	1.52	0.7	1.575x1.32 8 = 2.1	1.65	0.825	1.65

5.2.2. Wind Parallel to tank:

Cf = 1.2 (Ref fig 7.2-16, SBC 301)

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	Tank Dia, D mm	Cf	Projected Area, A, m2	Wind Force, kN	Force on each pier, kN	Axial on Pedestal Top, kNm
49FE-TNK-003	2680	1.2	$3.14x2.68^2/4$ = 5.68	7.6	3.8	+/- 3
49FE-TNK-006	2934	1.2	$3.14x2.934^2/4$ = 6.76	9.0	4.5	+/- 2.5
49FE-TNK-009	1328	1.2	$3.14x1.328^2/4$ = 1.4	1.9	0.95	+/- 3.6

5.2.3. Wind on Platform:

(a) Wind perpendicular to tank, Wf3 = (1.3 x 1.8 x 9 x 1.1) kN = 23.166 kN

Thus, Shear on each pier = 23.166/2 kN = 11.6 kN

Moment on pedestal top = (11.6 x 4) kN.m = 46.33 kN.m

(b) Wind parallel to tank, Wf4 = (1.3 x 1.8 x 1.1 x 2) kN = 5.15 kN

Hence, Shear on each pier = 5.15/2 kN = 2.575 kN

Push/Pull on pedestal top = +/- (2.575 x 4/7.6) kN = +/- 1.72 kN

Platform height = 1.1 m

Platform size = 9m x 2m

Cf = 1.8

5.3. Seismic Load(E)

The basic parameters for calculation of seismic loads is shown below:

Importance Factor, I = 1.0(As per SBC-301, Section-9.5)

Design Spectral Response Acceleration (Refer Clause-4.2.7b, MD-216-1000-EG-ST- CRT-1001)

Ss = 0.1g (Spectral Response Acceleration 0.2 second)

S1 = 0.02g (Spectral Response Acceleration 1 second)

S_{D_1} = 2/3*Fv*S1 = 2/3 * 2.4 * 0.02 = 0.032 (As per SBC-301, Section-9.4.4)

S_{DS} = 2/3*Fa*Ss = 2/3 * 1.6 * 0.1 = 0.106 (As per SBC-301, Section-9.4.4)

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Where S_{D1} Damped Design Spectral Response Acceleration@1 Sec

S_{DS} Damped Design Spectral Response Acceleration@0.2 Sec

Site Class = D

Site Coefficient, Fa = 1.60 (As per SBC-301, Table 9.4.3a)

Site Coefficient, Fv = 2.40 (As per SBC-301, Table 9.4.3b)

Building period Coefficient, Ct=0.04 (As per SBC-301, Table 10.9.3.2)

Hence, seismic base shear Vf = (1.2 x S_{DS} x W)/R = (1.2 x 0.106 x 407/3) kN = 17.3 kN

	Empty Load, De (kN)	Operating Load, Do (kN)	S_{DS}	Base Shear in Empty condition	Base Shear in Operating condition
49FE-TNK-003	133.6	530.15	0.106	14.2 kN	56.2 kN
49FE-TNK-006	174.5	725.75	0.106	18.5 kN	77 kN
49FE-TNK-009	15.3	26.5	0.106	1.65 kN	2.8 kN

For seismic in perpendicular tank direction a moment and for parallel to tank direction a vertical push/pull force has been considered.

6. STEEL BEARING PLATE & PEDESTAL SIZING:

	49FE-TNK-003	49FE-TNK-006	49FE-TNK-009
Bearing plate width	550 mm	550 mm	250 mm
Bearing plate length	2755 mm	3000 mm	1500 mm

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Pier width	**650 mm**	**650 mm**	**350 mm**
Pier length	**2855 mm**	**3100 mm**	**1600 mm**

Bearing plate width = saddle width +2x50mm

Bearing plate length = Saddle length + 2x50mm

Pier length = bearing plate length + 2x50mm

Pier width = bearing plate width + 2x50mm

7. DESIGN COMBINATIONS

Service Loads (Allowable Stress Design) (SBC-301, Section 2.4.1)

D

D + T

D +T+LL

D +0.75LL……*not used as non-critical*

D + (W or 0.7E)

D + 0.75W + 0.75LL

0.6D + W

0.6D + 0.7E

Factored loads (Strength Design) (SBC-301, Section 2.3.2)

1.4D

1.4D+1.4T

1.2(D + T) + 0.5LL……*not used as non-critical*

1.2D+ 1.2T + 1.6LL

1.2D + 1.6W + 1.0LL

1.2D + 1.0E

0.9D + 1.6W

0.9D + 1.0E

Where:

D = Dead Load, separate load case created for empty, operating and test condition as applicable.

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Lr = Live Load

W = Wind Load

E = Seismic Load

T = Temperature Load

8. DESIGN CALCULATION FOR FOUNDATION

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8.1 Design calculation of Footing for 49FE-TNK-003

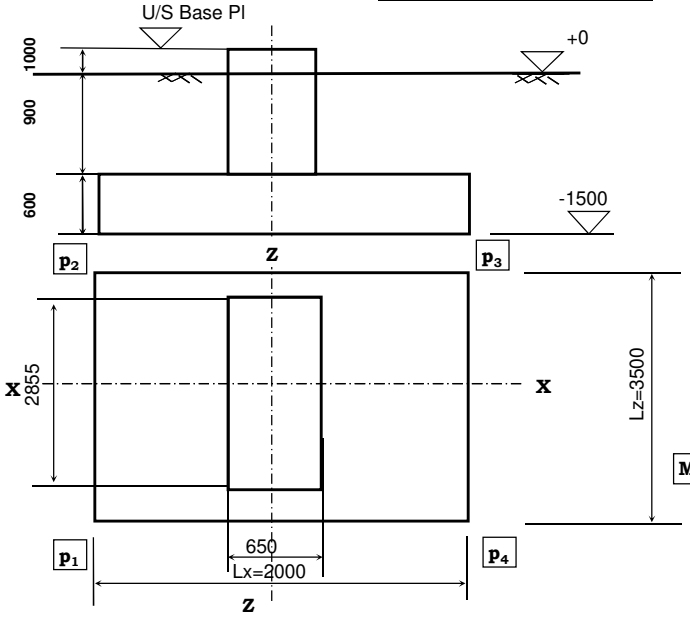
	Along-X	Along-Z
Size of column :	0.65m	2.86m
Size of fdn. raft :	2.00m	3.50m
Thickness of raft :	0.60 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-1.50 m	
U/S level Base Plate :	1.00 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

Calculation of Weight of Foundation :

Weight of raft & Col.= (2*3.5*0.6*24)+(0.65*2.855*1.9*24) = 185.42KN

Weight of soil = ((2*3.5)-(0.65*2.855))*0.9*18 = 83.34KN

Total = 268.76KN



Design Input:

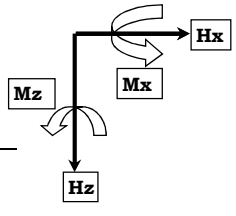
Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provided along Z direction at bottom=	16mm
Mininum percentage of reinforcement at bottom=	0.18%

Effective depth of slab along X direction = 542mm = 0.542m

Effective depth of slab along Z direction = 526mm = 0.526m

Clear projection along Z direction = 0.3225m

Clear projection along X direction = 0.675m



H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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(Refer attached load data dwg.)

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)
1	Empty Load (De)	67.00	0.00	0.0	0.0	12.0
2	Operating Load (Do)	266.00	0.00	0.0	0.0	48.0
3	Test Load (Dt)	266.00	0.00	0.0	0.0	48.0
4	Temparature Load (T)	0.00	107.00	0.0	0.0	0.0
5	Wind Long (WL)	5.00	7.50	0.0	0.0	0.0
6	Wind Transverse (WT)	0.00	0.00	0.0	22.0	67.0
7	Seismic Long (EL)	20.00	40.00	0.0	0.0	0.0
8	Seismic Transverse (ET)	0.00	0.00	0.0	40.0	100.0
9	Live Load (LL)	70.00	0.00	0.0	0.0	0.0

Combined Load Cases:		**Load at top of foundation**					**Load at bottom of foundation**				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
101	Do+T	266.0	107.0	0.0	0.0	48.0	534.76	107.00	267.50	0.00	48.00
102	Dt+T	266.0	107.0	0.0	0.0	48.0	534.76	107.00	267.50	0.00	48.00
103	Do+WL	271.0	7.5	0.0	0.0	48.0	539.76	7.50	18.75	0.00	48.00
104	Dt+WL	271.0	7.5	0.0	0.0	48.0	539.76	7.50	18.75	0.00	48.00
105	Do+WT	266.0	0.0	0.0	22.0	115.0	534.76	0.00	0.00	22.00	170.00
106	Dt+WT	266.0	0.0	0.0	22.0	115.0	534.76	0.00	0.00	22.00	170.00
107	Do+0.7EL	280.0	28.0	0.0	0.0	48.0	548.76	28.00	70.00	0.00	48.00
108	Dt+0.7EL	280.0	28.0	0.0	0.0	48.0	548.76	28.00	70.00	0.00	48.00
109	Do+0.7ET	266.0	0.0	0.0	28.0	118.0	534.76	0.00	0.00	28.00	188.00
110	Dt+0.7ET	266.0	0.0	0.0	28.0	118.0	534.76	0.00	0.00	28.00	188.00
111	0.6De+WL	35.2	7.5	0.0	0.0	7.2	303.96	7.50	18.75	0.00	7.20
112	0.6De+WT	40.2	0.0	0.0	22.0	74.2	308.96	0.00	0.00	22.00	129.20
113	0.6De+EL	20.2	40.0	0.0	0.0	7.2	288.96	40.00	100.00	0.00	7.20
114	0.6De+ET	40.2	0.0	0.0	40.0	107.2	308.96	0.00	0.00	40.00	207.20
115	Do+LL+T	336.0	107.0	0.0	0.0	48.0	604.76	107.00	267.50	0.00	48.00
116	Do+0.75WL+0.75LL	322.3	5.6	0.0	0.0	48.0	591.01	5.63	14.06	0.00	48.00
117	Do+0.75WT+0.75LL	318.5	0.0	0.0	16.5	98.3	587.26	0.00	0.00	16.50	139.50

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Ultimate Conditions											
201	1.4(Do+T)	372.4	149.8	0.0	0.0	67.2	641.16	149.80	374.50	0.00	67.20
202	1.4(Dt+T)	372.4	149.8	0.0	0.0	67.2	641.16	149.80	374.50	0.00	67.20
203	1.2Do+1.6WL	327.2	12.0	0.0	0.0	57.6	595.96	12.00	30.00	0.00	57.60
204	1.2Dt+1.6WL	327.2	12.0	0.0	0.0	57.6	595.96	12.00	30.00	0.00	57.60
205	1.2Do+1.6WT	319.2	0.0	0.0	35.2	164.8	587.96	0.00	0.00	35.20	252.80
206	1.2Dt+1.6WT	319.2	0.0	0.0	35.2	164.8	587.96	0.00	0.00	35.20	252.80
207	1.2Do+EL	339.2	40.0	0.0	0.0	57.6	607.96	40.00	100.00	0.00	57.60
208	1.2Dt+EL	339.2	40.0	0.0	0.0	57.6	607.96	40.00	100.00	0.00	57.60
209	1.2Do+ET	319.2	0.0	0.0	40.0	157.6	587.96	0.00	0.00	40.00	257.60
210	1.2Dt+ET	319.2	0.0	0.0	40.0	157.6	587.96	0.00	0.00	40.00	257.60
211	0.9De+1.6WL	52.3	12.0	0.0	0.0	10.8	321.06	12.00	30.00	0.00	10.80
212	0.9De+1.6WT	60.3	0.0	0.0	40.0	110.8	329.06	0.00	0.00	40.00	210.80
213	0.9De+EL	40.30	40.00	0.00	0.00	10.80	309.06	40.00	100.00	0.00	10.80
214	0.9De+ET	60.30	0.00	0.00	40.00	110.80	329.06	0.00	0.00	40.00	210.80
215	1.2Do+1.2T+1.6LL	431.20	128.40	0.00	0.00	57.60	699.96	128.40	321.00	0.00	57.60
216	1.2Do+1.6WL+0LL	397.20	12.00	0.00	0.00	57.60	665.96	12.00	30.00	0.00	57.60
217	1.2Do+1.6WT+1.0LL	389.20	0.00	0.00	35.20	164.80	657.96	0.00	0.00	35.20	252.80

Check for overturning and sliding:

	Load Cases:	Load at b/o fdn					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
101	Do+T	534.76	107.00	267.50	0.00	0.00	2.00	NA	2.00	NA
102	Dt+T	534.76	107.00	267.50	0.00	1.00	2.00	535	2.00	NA
103	Do+WL	539.76	7.50	18.75	0.00	2.00	28.79	270	28.79	NA
104	Dt+WL	539.76	7.50	18.75	0.00	3.00	28.79	180	28.79	NA
105	Do+WT	534.76	0.00	0.00	22.00	4.00	NA	134	NA	9.72
106	Dt+WT	534.76	0.00	0.00	22.00	5.00	NA	107	NA	9.72
107	Do+0.7EL	548.76	28.00	70.00	0.00	6.00	7.84	91	7.84	NA
108	Dt+0.7EL	548.76	28.00	70.00	0.00	7.00	7.84	78	7.84	NA
109	Do+0.7ET	534.76	0.00	0.00	28.00	8.00	NA	67	NA	7.64
110	Dt+0.7ET	534.76	0.00	0.00	28.00	9.00	NA	59	NA	7.64
111	0.6De+WL	303.96	7.50	18.75	0.00	10.00	16.21	30	16.21	NA
112	0.6De+WT	308.96	0.00	0.00	22.00	11.00	NA	28	NA	5.62
113	0.6De+EL	288.96	40.00	100.00	0.00	12.00	2.89	24	2.89	NA
114	0.6De+ET	308.96	0.00	0.00	40.00	13.00	NA	24	NA	3.09
115	Do+LL+T	604.76	107.00	267.50	0.00	14.00	2.26	43	2.26	NA
116	Do+0.75WL+0.75LL	591.01	5.63	14.06	0.00	15.00	42.03	39	42.03	NA
117	Do+0.75WT+0.75LL	587.26	0.00	0.00	16.50	16.00	NA	37	NA	14.24
						Min.=	2.0	23.8	2.0	3.1

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Load Combination & Calculation of Base pressure

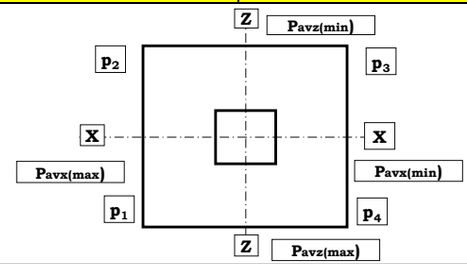
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 20 kN/m^2

Net SBC of soil for DL, LL= 350 kN/m^2 For WL/SL case = 437.5 kN/m^2

Gross SBC of soil for DL, LL= 377 kN/m^2 For WL/SL case = 464.5 kN/m^2



Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m^2)												
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
101	Do+T	534.76	267.50	48.00	76.39	114.64	11.76	250.00	250.00	0.00	0.00	223.00	250.00	0.00	125.00	125.00	O.K
102	Dt+T	534.76	267.50	48.00	76.39	114.64	11.76	250.00	250.00	0.00	0.00	223.00	250.00	0.00	125.00	125.00	O.K
103	Do+WL	539.76	18.75	48.00	77.11	8.04	11.76	96.90	73.39	80.83	57.32	69.90	85.15	69.08	88.87	65.36	O.K
104	Dt+WL	539.76	18.75	48.00	77.11	8.04	11.76	96.90	73.39	80.83	57.32	69.90	85.15	69.08	88.87	65.36	O.K
105	Do+WT	534.76	0.00	170.00	76.39	0.00	41.63	118.03	34.76	118.03	34.76	91.03	76.40	76.40	118.03	34.76	O.K
106	Dt+WT	534.76	0.00	170.00	76.39	0.00	41.63	118.03	34.76	118.03	34.76	91.03	76.40	76.40	118.03	34.76	O.K
107	Do+0.7EL	548.76	70.00	48.00	78.39	30.00	11.76	120.15	96.64	60.15	36.64	93.15	108.40	48.40	90.15	66.64	O.K
108	Dt+0.7EL	548.76	70.00	48.00	78.39	30.00	11.76	120.15	96.64	60.15	36.64	93.15	108.40	48.40	90.15	66.64	O.K
109	Do+0.7ET	534.76	0.00	188.00	76.39	0.00	46.04	122.44	30.35	122.44	30.35	95.44	76.40	76.40	122.44	30.35	O.K
110	Dt+0.7ET	534.76	0.00	188.00	76.39	0.00	46.04	122.44	30.35	122.44	30.35	95.44	76.40	76.40	122.44	30.35	O.K
111	0.6De+WL	303.96	18.75	7.20	43.42	8.04	1.76	53.22	49.70	37.15	33.62	26.22	51.46	35.39	45.19	41.66	O.K
112	0.6De+WT	308.96	0.00	129.20	44.14	0.00	31.64	75.78	12.50	75.78	12.50	48.78	44.14	44.14	75.78	12.50	O.K
113	0.6De+EL	288.96	100.00	7.20	41.28	42.86	1.76	100.00	100.00	0.00	0.00	73.00	100.00	0.00	50.00	50.00	O.K
114	0.6De+ET	308.96	0.00	207.20	44.14	0.00	50.74	100.00	100.00	0.00	0.00	73.00	100.00	0.00	50.00	50.00	O.K
115	Do+LL+T	604.76	267.50	48.00	86.39	114.64	11.76	255.00	255.00	0.00	0.00	228.00	255.00	0.00	127.50	127.50	O.K
116	Do+0.75WL+0.75LL	591.01	14.06	48.00	84.43	6.03	11.76	102.21	78.70	90.16	66.65	75.21	90.46	78.41	96.19	72.68	O.K
117	Do+0.75WT+0.75LL	587.26	0.00	139.50	83.89	0.00	34.16	118.06	49.73	118.06	49.73	91.06	83.90	83.90	118.06	49.73	O.K

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201	1.4(Do+T)	641.16	374.50	67.20	91.59	160.50	16.46	420.00	420.00	0.00	0.00	393.00	420.00	0.00	210.00	210.00
202	1.4(Dt+T)	641.16	374.50	67.20	91.59	160.50	16.46	420.00	420.00	0.00	0.00	393.00	420.00	0.00	210.00	210.00
203	1.2Do+1.6WL	595.96	30.00	57.60	85.14	12.86	14.11	112.10	83.89	86.39	58.17	85.10	98.00	72.28	99.25	71.03
204	1.2Dt+1.6WL	595.96	30.00	57.60	85.14	12.86	14.11	112.10	83.89	86.39	58.17	85.10	98.00	72.28	99.25	71.03
205	1.2Do+1.6WT	587.96	0.00	252.80	83.99	0.00	61.91	145.90	22.08	145.90	22.08	118.90	83.99	83.99	145.90	22.08
206	1.2Dt+1.6WT	587.96	0.00	252.80	83.99	0.00	61.91	145.90	22.08	145.90	22.08	118.90	83.99	83.99	145.90	22.08
207	1.2Do+EL	607.96	100.00	57.60	86.85	42.86	14.11	143.81	115.60	58.10	29.89	116.81	129.71	44.00	100.96	72.75
208	1.2Dt+EL	607.96	100.00	57.60	86.85	42.86	14.11	143.81	115.60	58.10	29.89	116.81	129.71	44.00	100.96	72.75
209	1.2Do+ET	587.96	0.00	257.60	83.99	0.00	63.09	147.08	20.91	147.08	20.91	120.08	84.00	84.00	147.08	20.91
210	1.2Dt+ET	587.96	0.00	257.60	83.99	0.00	63.09	147.08	20.91	147.08	20.91	120.08	84.00	84.00	147.08	20.91
211	0.9De+1.6WL	321.06	30.00	10.80	45.87	12.86	2.64	61.37	56.08	35.65	30.36	34.37	58.73	33.01	48.51	43.22
212	0.9De+1.6WT	329.06	0.00	210.80	47.01	0.00	51.62	98.63	-4.62	98.63	-4.62	71.63	47.01	47.01	98.63	-4.62
213	0.9De+EL	309.06	100.00	10.80	44.15	42.86	2.64	89.65	84.36	3.94	-1.35	62.65	87.01	1.30	46.80	41.51
214	0.9De+ET	329.06	0.00	210.80	47.01	0.00	51.62	98.63	-4.62	98.63	-4.62	71.63	47.01	47.01	98.63	-4.62
215	1.2Do+1.2T+1.6LL	699.96	321.00	57.60	99.99	137.57	14.11	360.00	360.00	0.00	0.00	333.00	360.00	0.00	180.00	180.00
216	1.2Do+1.6WL+0LL	665.96	30.00	57.60	95.14	12.86	14.11	122.10	93.89	96.39	68.17	95.10	108.00	82.28	109.25	81.03
217	1.2Do+1.6WT+1.0LL	657.96	0.00	252.80	93.99	0.00	61.91	155.90	32.08	155.90	32.08	128.90	93.99	93.99	155.90	32.08
	Max :	700	375	258												
	Min :	289	0	7												

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 223.00 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 243.00 kN/m^2

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 393.00 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 413.00 kN/m^2

Sample calculation of values of base pressure:

Load case : Dt+T

Maximum pressure, p_1 (pmax) = 76.39+114.64+11.76= 202.79 kN/m^2

p_2= 76.39+114.64-11.76= 179.28 kN/m^2

p_3= 76.39-(114.64+11.76)= -50.00 kN/m^2

p_4= 76.39-114.64+11.76= -26.49 kN/m^2

Maxm pressure excluding weight of soil (p_{max}-γh)= 250.00-1.5*18= 223.00 kN/m^2

Maximum average Pressure along X, $p_{avx(max)}$ = (250.00+250.00)/2= 191.04 kN/m^2

Minimum average Pressure along X, $p_{avx(minx)}$ = (0.00+0.00)/2= -38.25 kN/m^2

Maximum average Pressure along Y, $p_{avy(max)}$ = (250.00+0.00)/2= 88.15 kN/m^2

Minimum average Pressure along Y, $p_{avy(minx)}$ = (0.00+250.00)/2= 64.64 kN/m^2

Max PressureDL+LL	270.00	**OK**
Max Pressure Including SL/WL	275.00	**OK**

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Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.65+2*0.5*0.526)+(2.855+2*0.5*0.526))*2

= 9.114 m

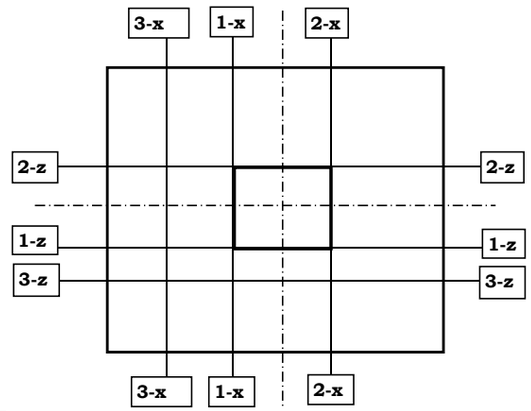
Punching Area= ((0.65+2*0.5*0.526)*(2.855+2*0.5*0.526))

= 3.98 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.6*24+0.9*18)

= 30.60 kN/m^2

Area of footing, A_f= 7.00 m^2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	B.M_{1x} kNm	B.M_{2x} kNm	S.F_{3x} kN	S.$F_{(punching)}$ kN
Ultimate Condition								
201	1.4(Do+T)	278.25	141.75	392.07	77.95	3.79	49.93	184.20
202	1.4(Dt+T)	278.25	141.75	392.07	77.95	3.79	49.93	184.20
203	1.2Do+1.6WL	89.32	80.96	96.29	14.70	10.15	8.85	164.70
204	1.2Dt+1.6WL	89.32	80.96	96.29	14.70	10.15	8.85	164.70
205	1.2Do+1.6WT	83.99	83.99	83.99	12.16	12.16	7.10	161.25
206	1.2Dt+1.6WT	83.99	83.99	83.99	12.16	12.16	7.10	161.25
207	1.2Do+EL	100.78	72.93	124.01	20.38	5.25	12.80	169.88
208	1.2Dt+EL	100.78	72.93	124.01	20.38	5.25	12.80	169.88
209	1.2Do+ET	84.00	84.00	84.00	12.17	12.17	7.10	161.25
210	1.2Dt+ET	84.00	84.00	84.00	12.17	12.17	7.10	161.25
211	0.9De+1.6WL	50.05	41.69	57.02	5.75	1.21	3.63	46.10
212	0.9De+1.6WT	47.01	47.01	47.01	3.74	3.74	2.18	49.55
213	0.9De+EL	58.08	30.23	81.31	10.65	-4.48	7.12	40.93
214	0.9De+ET	47.01	47.01	47.01	3.74	3.74	2.18	49.55
215	1.2Do+1.2T+1.6LL	238.50	121.50	336.06	65.82	2.26	42.22	209.57
216	1.2Do+1.6WL+0LL	99.32	90.96	106.29	16.97	12.43	10.18	194.90
217	1.2Do+1.6WT+1.0LL	93.99	93.99	93.99	14.44	14.44	8.43	191.45

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Sample calculation of BM & SF

Load case : 201

p_{1x} =	420-(420-0)*0.675/2 =	278.25 kN/m^2	Overhang =	0.675 m
p_{2x} =	0+(420-0)*0.675/2 =	141.75 kN/m^2	Effective depth =	0.542 m
p_{3x} =	420-(420-0)*(0.675-0.542)/2 =	392.07 kN/m^2		

BM_{1x} =	(278.25-30.6)*0.675*0.675/2+0.5*(420-278.25)*0.675*0.675*0.67 =	77.95 kN-m/m
BM_{2x} =	(141.75-30.6)*0.675*0.675/2-0.5*(-0+141.75)*0.675*0.675*0.67 =	3.79 kN-m/m
SF_{3x} =	(392.07-30.6)*(0.675-0.542)+0.5*(420-392.07)*(0.675-0.542) =	49.93 kN/m
$S.F_{(punching)}$ =	(7-3.98)*(91.59-30.6) =	184.20 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	77.95	0.27	0.35	0.001	459.29	975.60	16ϕT	206.09	300	200	1005.31
At top	-4.48	-0.02	0.02	0.000	26.24	975.60	16ϕT	206.09	300	200	1005.31

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, $(\phi\ Vc)/bd$ = 0.740 N/mm2

Shear at a distance 'd' from face of column, Vu= 49.93 kN/m

Shear stress , tv= (49.93*1000)/(1000*542)

= 0.09 N/mm2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu=	184.20 kN	
Strength of Concrete in shear Vc1 =	47657.89 kN	Table 22.6.5.2_ (a)
Strength of Concrete in shear Vc2 =	7767.25 kN	Table 22.6.5.2_ (b)
Strength of Concrete in shear Vc3 =	9741.39 kN	Table 22.6.5.2_ (c)
Design Vc=	7767.25 kN	
Allowable shear stress, $(\phi\ Vc)/bd$ =	1.18 N/mm2	

Design punching shear stress, τ = (184.2*1000)/(9114*542)

= 0.04 N/mm^2 **No shear reinforcement reqd.**

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Calculation of Bending Moment and Shear Force in Z direction :

p_{1z} =pressure at the face of column at section 1z-1z in kN/m^2.
p_{2z} =pressure at the face of column at section 2z-2z in kN/m^2.
p_{3z} =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
$B.M_{1z}$ =Moment at the face of column at section 1z-1z in kN-m/m.
$B.M_{2z}$ =Moment at the face of column at section 2z-2z in kN-m/m.
$S.F_{3z}$ = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions							
201	1.4(Do+T)	210.00	210.00	210.00	9.33	9.33	-36.51
202	1.4(Dt+T)	210.00	210.00	210.00	9.33	9.33	-36.51
203	1.2Do+1.6WL	96.65	73.63	100.89	3.53	2.15	-14.14
204	1.2Dt+1.6WL	96.65	73.63	100.89	3.53	2.15	-14.14
205	1.2Do+1.6WT	134.49	33.49	153.10	5.80	-0.25	-24.20
206	1.2Dt+1.6WT	134.49	33.49	153.10	5.80	-0.25	-24.20
207	1.2Do+EL	98.36	75.35	102.60	3.61	2.24	-14.49
208	1.2Dt+EL	98.36	75.35	102.60	3.61	2.24	-14.49
209	1.2Do+ET	135.45	32.54	154.42	5.86	-0.30	-24.45
210	1.2Dt+ET	135.45	32.54	154.42	5.86	-0.30	-24.45
211	0.9De+1.6WL	48.02	43.71	48.82	0.92	0.66	-3.68
212	0.9De+1.6WT	89.12	4.89	104.63	3.37	-1.67	-14.45
213	0.9De+EL	46.31	42.00	47.11	0.83	0.58	-3.33
214	0.9De+ET	89.12	4.89	104.63	3.37	-1.67	-14.45
215	1.2Do+1.2T+1.6LL	180.00	180.00	180.00	7.77	7.77	-30.40
216	1.2Do+1.6WL+0LL	106.65	83.63	110.89	4.05	2.67	-16.17
217	1.2Do+1.6WT+1.0LL	144.49	43.49	163.10	6.32	0.27	-26.23

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Sample calculation of BM & SF

Load case : 201

p_{1z} = 210-(210-210)*0.3225/3.5= 210.00 kN/m^2

p_{2z} = 210+(210-210)*0.3225/3.5= 210.00 kN/m^2

p_{3z} = 210-(210-210)*(0.3225-0.526)/3.5= 210.00 kN/m^2

Overhang = 0.3225 m

Effective depth = 0.526 m

BM_{1z} = (210.00-30.6)*0.3225*0.3225/2+0.5*(210-210.00)*0.3225*0.3225*0.67 = 9.33 kN-m/m

BM_{2z} = (210.00-30.6)*0.3225*0.3225/2-0.5*(-210+210.00)*0.3225*0.3225*0.67 = 9.33 kN-m/m

SF_{3z} = (210.00-30.6)*(0.3225-0.526)+0.5*(210-210.00)*(0.3225-0.526) = -36.51 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	9.33	0.034	0.045	0.0001	58.06	946.80	16ϕT	212.36	300	200	1005.3
At top	-1.67	-0.006	-0.008	0.0000	-0.10	946.80	16ϕT	212.36	300	200	1005.3

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm2

Shear at a distance 'd' from face of column, V_u= -14.14 kN/m

Shear stress , τ_v= (-14.14*1000)/(1000*526)

= -0.03 N/mm2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				
	At Bottom	16ϕT	200	c/c
	At Top	16ϕT	200	c/c
Along Z Direction :				
	At Bottom	16ϕT	200	c/c
	At Top	16ϕT	200	c/c

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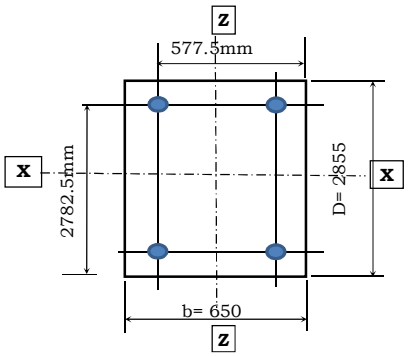
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8.2 Design calculation of cOLUMN for 49FE-TNK-003

COLUMN MKD C1

Input:

Clear cover to reinf.=	50mm	Table 20.6.1.3.1
Dia. of main reinf.=	25mm	
Dia. of link=	10mm	
Effective cover =	72.5mm	
Grade of Concrete (f_c') =	M35	
Grade of Steel (f_y) =	Fy420	
Grade of Reinf(links).=	Fy420	
Min. Steel Ratio	0.15%	
Max. Steel Ratio	4.00%	
Self weight of Column =	80.17kN	



For Compression-controlled sections
Strength reduction factor ϕ= 0.65 Table 21.2.2

Unsupproted length of column about X (l_{ux}) =	1800mm
Unsupproted length of column about Z (l_{uz}) =	1800mm

For Columns not braced against sideway, Where, Effective length factor K= 2

Slenderness ratio,	K l_{ux} / r_x =	4.20	< 22 SECTION 6.2.5a	Radius of gyration r_x=	857	mm
	K l_{uz} / r_z =	18.46	< 22 SECTION 6.2.5a	Radius of gyration r_z=	195	mm

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γ_y = 0.96

γ_z = 0.81

Load Case	Load Combination	Factored Load From STAAD			Load at Column Base			Nominal Load/Moment			Kn = Pn/f'_cbD (N/mm^2)
		Pu (kN)	Hux (kN)	Huz (kN)	Pu (kN)	Muz (kNm)	Mux (kNm)	Pn (kN)	Mnz (kNm)	Mnx (kNm)	
201	1.4(Do+T)	372.40	149.8	0.0	452.57	269.6	0.0	696.3	414.8	0.0	0.011
202	1.4(Dt+T)	372.40	149.8	0.0	452.57	269.6	0.0	696.3	414.8	0.0	0.011
203	1.2Do+1.6WL	327.20	12.0	0.0	407.37	21.6	0.0	626.7	33.2	0.0	0.010
204	1.2Dt+1.6WL	327.20	12.0	0.0	407.37	21.6	0.0	626.7	33.2	0.0	0.010
205	1.2Do+1.6WT	319.20	0.0	35.2	399.37	0.0	63.4	614.4	0.0	97.5	0.009
206	1.2Dt+1.6WT	319.20	0.0	35.2	399.37	0.0	63.4	614.4	0.0	97.5	0.009
207	1.2Do+EL	339.20	40.0	0.0	419.37	72.0	0.0	645.2	110.8	0.0	0.010
208	1.2Dt+EL	339.20	40.0	0.0	419.37	72.0	0.0	645.2	110.8	0.0	0.010
209	1.2Do+ET	319.20	0.0	40.0	399.37	0.0	72.0	614.4	0.0	110.8	0.009
210	1.2Dt+ET	319.20	0.0	40.0	399.37	0.0	72.0	614.4	0.0	110.8	0.009
211	0.9De+1.6WL	52.30	12.0	0.0	132.47	21.6	0.0	203.8	33.2	0.0	0.003
212	0.9De+1.6WT	60.30	0.0	40.0	140.47	0.0	72.0	216.1	0.0	110.8	0.003
213	0.9De+EL	40.30	40.0	0.0	120.47	72.0	0.0	185.3	110.8	0.0	0.003
214	0.9De+ET	60.30	0.0	40.0	140.47	0.0	72.0	216.1	0.0	110.8	0.003
215	1.2Do+1.2T+1.6LL	431.20	128.4	0.0	511.37	231.1	0.0	786.7	355.6	0.0	0.012
216	1.2Do+1.6WL+0LL	397.20	12.0	0.0	477.37	21.6	0.0	734.4	33.2	0.0	0.011
217	1.2Do+1.6WT+1.0LL	389.20	0.0	35.2	469.37	0.0	63.4	722.1	0.0	97.5	0.011

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Reinforcement: Provide,

4-Nos	25 mm Dia	1964.29 Sq mm	(At Corner)
22-Nos	25 mm Dia	10803.57 Sq mm	(Along Z axis)
4-Nos	25 mm Dia	1964.29 Sq mm	(Along Y axis)

A_{sc} pvd. = 14732.14 Sq mm

Total A_{sc} = 0.79 % ρ_g = 0.008

β = P_n / P_0 (range is 0.55 to 0.70) Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 6095.778 T SECTION 22.4.2.2

Load Comb	From Chart of SP17(M)-R35-420.8		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f_cb^2D$	$R_{nx}=M_{nox}/f_cb^2D$						
1.4(Do+T)	0.050	0.050	2110.9	9271.8	0.55	1.16	0.15	O.K
1.4(Dt+T)	0.050	1.050	2110.9	194707.6	0.55	1.16	0.15	O.K
1.2Do+1.6WL	0.050	2.050	2110.9	380143.4	0.55	1.16	0.01	O.K
1.2Dt+1.6WL	0.050	3.050	2110.9	565579.2	0.55	1.16	0.01	O.K
1.2Do+1.6WT	0.050	4.050	2110.9	751015.1	0.55	1.16	0.00	O.K
1.2Dt+1.6WT	0.050	5.050	2110.9	936450.9	0.55	1.16	0.00	O.K
1.2Do+EL	0.050	6.050	2110.9	1121886.7	0.55	1.16	0.03	O.K
1.2Dt+EL	0.050	7.050	2110.9	1307322.5	0.55	1.16	0.03	O.K
1.2Do+ET	0.050	8.050	2110.9	1492758.3	0.55	1.16	0.00	O.K
1.2Dt+ET	0.050	9.050	2110.9	1678194.2	0.55	1.16	0.00	O.K
0.9De+1.6WL	0.050	10.050	2110.9	1863630.0	0.55	1.16	0.01	O.K
0.9De+1.6WT	0.050	11.050	2110.9	2049065.8	0.55	1.16	0.00	O.K
0.9De+EL	0.050	12.050	2110.9	2234501.6	0.55	1.16	0.03	O.K
0.9De+ET	0.050	13.050	2110.9	2419937.4	0.55	1.16	0.00	O.K
1.2Do+1.2T+1.6LL	0.050	14.050	2110.9	2605373.3	0.55	1.16	0.13	O.K
1.2Do+1.6WL+0LL	0.050	15.050	2110.9	2790809.1	0.55	1.16	0.01	O.K
1.2Do+1.6WT+1.0LL	0.050	16.050	2110.9	2976244.9	0.55	1.16	0.00	O.K

Links, 10 mm dia @ 250 C/C

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8.3 Design calculation of Footing of 49FE-TNK-006

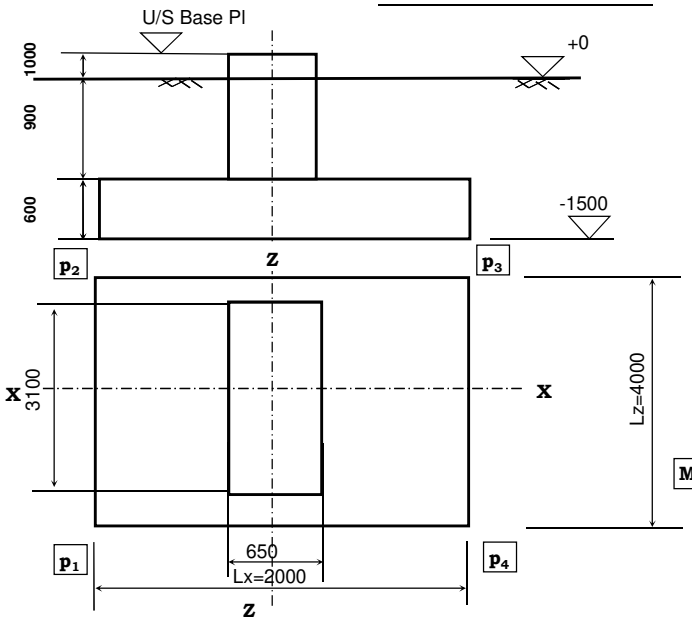
	Along-X	Along-Z
Size of column :	0.65m	3.10m
Size of fdn. raft :	2.00m	4.00m
Thickness of raft :	0.60 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-1.50 m	
U/S level Base Plate :	1.00 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

Calculation of Weight of Foundation :

Weight of raft & Col.= (2*4*0.6*24)+(0.65*3.1*1.9*24) = 207.08KN

Weight of soil = ((2*4)-(0.65*3.1))*0.9*18 = 96.96KN

Total = 304.04KN



Design Input:

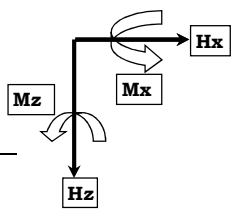
Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provided along Z direction at bottom=	16mm
Mininum percentage of reinforcement at bottom=	0.18%

Effective depth of slab along X direction = 542mm = 0.542m

Effective depth of slab along Z direction = 526mm = 0.526m

Clear projection along Z direction = 0.45m

Clear projection along X direction = 0.675m



H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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(Refer attached load data dwg.)

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)
1	Empty Load (De)	87.50	0.00	0.0	0.0	12.0
2	Operating Load (Do)	375.00	0.00	0.0	0.0	70.0
3	Test Load (Dt)	375.00	0.00	0.0	0.0	70.0
4	Temparature Load (T)	0.00	146.00	0.0	0.0	0.0
5	Wind Long (WL)	5.00	7.50	0.0	0.0	0.0
6	Wind Transverse (WT)	0.00	0.00	0.0	22.0	67.0
7	Seismic Long (EL)	20.00	40.00	0.0	0.0	0.0
8	Seismic Transverse (ET)	0.00	0.00	0.0	40.0	100.0
9	Live Load (LL)	70.00	0.00	0.0	0.0	0.0

Combined Load Cases:		**Load at top of foundation**					**Load at bottom of foundation**				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
101	Do+T	375.0	146.0	0.0	0.0	70.0	679.04	146.00	365.00	0.00	70.00
102	Dt+T	375.0	146.0	0.0	0.0	70.0	679.04	146.00	365.00	0.00	70.00
103	Do+WL	380.0	7.5	0.0	0.0	70.0	684.04	7.50	18.75	0.00	70.00
104	Dt+WL	380.0	7.5	0.0	0.0	70.0	684.04	7.50	18.75	0.00	70.00
105	Do+WT	375.0	0.0	0.0	22.0	137.0	679.04	0.00	0.00	22.00	192.00
106	Dt+WT	375.0	0.0	0.0	22.0	137.0	679.04	0.00	0.00	22.00	192.00
107	Do+0.7EL	389.0	28.0	0.0	0.0	70.0	693.04	28.00	70.00	0.00	70.00
108	Dt+0.7EL	389.0	28.0	0.0	0.0	70.0	693.04	28.00	70.00	0.00	70.00
109	Do+0.7ET	375.0	0.0	0.0	28.0	140.0	679.04	0.00	0.00	28.00	210.00
110	Dt+0.7ET	375.0	0.0	0.0	28.0	140.0	679.04	0.00	0.00	28.00	210.00
111	0.6De+WL	47.5	7.5	0.0	0.0	7.2	351.54	7.50	18.75	0.00	7.20
112	0.6De+WT	52.5	0.0	0.0	22.0	74.2	356.54	0.00	0.00	22.00	129.20
113	0.6De+EL	32.5	40.0	0.0	0.0	7.2	336.54	40.00	100.00	0.00	7.20
114	0.6De+ET	52.5	0.0	0.0	40.0	107.2	356.54	0.00	0.00	40.00	207.20
115	Do+LL+T	445.0	146.0	0.0	0.0	70.0	749.04	146.00	365.00	0.00	70.00
116	Do+0.75WL+0.75LL	431.3	5.6	0.0	0.0	70.0	735.29	5.63	14.06	0.00	70.00
117	Do+0.75WT+0.75LL	427.5	0.0	0.0	16.5	120.3	731.54	0.00	0.00	16.50	161.50

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Ultimate Conditions											
201	1.4(Do+T)	525.0	204.4	0.0	0.0	98.0	829.04	204.40	511.00	0.00	98.00
202	1.4(Dt+T)	525.0	204.4	0.0	0.0	98.0	829.04	204.40	511.00	0.00	98.00
203	1.2Do+1.6WL	458.0	12.0	0.0	0.0	84.0	762.04	12.00	30.00	0.00	84.00
204	1.2Dt+1.6WL	458.0	12.0	0.0	0.0	84.0	762.04	12.00	30.00	0.00	84.00
205	1.2Do+1.6WT	450.0	0.0	0.0	35.2	191.2	754.04	0.00	0.00	35.20	279.20
206	1.2Dt+1.6WT	450.0	0.0	0.0	35.2	191.2	754.04	0.00	0.00	35.20	279.20
207	1.2Do+EL	470.0	40.0	0.0	0.0	84.0	774.04	40.00	100.00	0.00	84.00
208	1.2Dt+EL	470.0	40.0	0.0	0.0	84.0	774.04	40.00	100.00	0.00	84.00
209	1.2Do+ET	450.0	0.0	0.0	40.0	184.0	754.04	0.00	0.00	40.00	284.00
210	1.2Dt+ET	450.0	0.0	0.0	40.0	184.0	754.04	0.00	0.00	40.00	284.00
211	0.9De+1.6WL	70.8	12.0	0.0	0.0	10.8	374.79	12.00	30.00	0.00	10.80
212	0.9De+1.6WT	78.8	0.0	0.0	40.0	110.8	382.79	0.00	0.00	40.00	210.80
213	0.9De+EL	58.75	40.00	0.00	0.00	10.80	362.79	40.00	100.00	0.00	10.80
214	0.9De+ET	78.75	0.00	0.00	40.00	110.80	382.79	0.00	0.00	40.00	210.80
215	1.2Do+1.2T+1.6LL	562.00	175.20	0.00	0.00	84.00	866.04	175.20	438.00	0.00	84.00
216	1.2Do+1.6WL+0LL	528.00	12.00	0.00	0.00	84.00	832.04	12.00	30.00	0.00	84.00
217	1.2Do+1.6WT+1.0LL	520.00	0.00	0.00	35.20	191.20	824.04	0.00	0.00	35.20	279.20

Check for overturning and sliding:

	Load Cases:	Load at b/o fdn					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
101	Do+T	679.04	146.00	365.00	0.00	0.00	1.86	NA	1.86	NA
102	Dt+T	679.04	146.00	365.00	0.00	1.00	1.86	679	1.86	NA
103	Do+WL	684.04	7.50	18.75	0.00	2.00	36.48	342	36.48	NA
104	Dt+WL	684.04	7.50	18.75	0.00	3.00	36.48	228	36.48	NA
105	Do+WT	679.04	0.00	0.00	22.00	4.00	NA	170	NA	12.35
106	Dt+WT	679.04	0.00	0.00	22.00	5.00	NA	136	NA	12.35
107	Do+0.7EL	693.04	28.00	70.00	0.00	6.00	9.90	116	9.90	NA
108	Dt+0.7EL	693.04	28.00	70.00	0.00	7.00	9.90	99	9.90	NA
109	Do+0.7ET	679.04	0.00	0.00	28.00	8.00	NA	85	NA	9.70
110	Dt+0.7ET	679.04	0.00	0.00	28.00	9.00	NA	75	NA	9.70
111	0.6De+WL	351.54	7.50	18.75	0.00	10.00	18.75	35	18.75	NA
112	0.6De+WT	356.54	0.00	0.00	22.00	11.00	NA	32	NA	6.48
113	0.6De+EL	336.54	40.00	100.00	0.00	12.00	3.37	28	3.37	NA
114	0.6De+ET	356.54	0.00	0.00	40.00	13.00	NA	27	NA	3.57
115	Do+LL+T	749.04	146.00	365.00	0.00	14.00	2.05	54	2.05	NA
116	Do+0.75WL+0.75LL	735.29	5.63	14.06	0.00	15.00	52.29	49	52.29	NA
117	Do+0.75WT+0.75LL	731.54	0.00	0.00	16.50	16.00	NA	46	NA	17.73
						Min.=	1.9	27.4	1.9	3.6

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Load Combination & Calculation of Base pressure

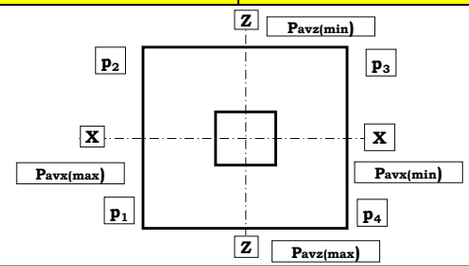
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 20 kN/m^2

Net SBC of soil for DL, LL= 350 kN/m^2 For WL/SL case = 437.5 kN/m^2

Gross SBC of soil for DL, LL= 377 kN/m^2 For WL/SL case = 464.5 kN/m^2



Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m^2)												Remarks
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	
101	Do+T	679.04	365.00	70.00	84.88	136.88	13.13	300.00	300.00	0.00	0.00	273.00	300.00	0.00	150.00	150.00	O.K
102	Dt+T	679.04	365.00	70.00	84.88	136.88	13.13	300.00	300.00	0.00	0.00	273.00	300.00	0.00	150.00	150.00	O.K
103	Do+WL	684.04	18.75	70.00	85.51	7.03	13.13	105.66	79.41	91.60	65.35	78.66	92.54	78.48	98.63	72.38	O.K
104	Dt+WL	684.04	18.75	70.00	85.51	7.03	13.13	105.66	79.41	91.60	65.35	78.66	92.54	78.48	98.63	72.38	O.K
105	Do+WT	679.04	0.00	192.00	84.88	0.00	36.00	120.88	48.88	120.88	48.88	93.88	84.88	84.88	120.88	48.88	O.K
106	Dt+WT	679.04	0.00	192.00	84.88	0.00	36.00	120.88	48.88	120.88	48.88	93.88	84.88	84.88	120.88	48.88	O.K
107	Do+0.7EL	693.04	70.00	70.00	86.63	26.25	13.13	126.01	99.76	73.51	47.26	99.01	112.89	60.39	99.76	73.51	O.K
108	Dt+0.7EL	693.04	70.00	70.00	86.63	26.25	13.13	126.01	99.76	73.51	47.26	99.01	112.89	60.39	99.76	73.51	O.K
109	Do+0.7ET	679.04	0.00	210.00	84.88	0.00	39.38	124.26	45.51	124.26	45.51	97.26	84.89	84.89	124.26	45.51	O.K
110	Dt+0.7ET	679.04	0.00	210.00	84.88	0.00	39.38	124.26	45.51	124.26	45.51	97.26	84.89	84.89	124.26	45.51	O.K
111	0.6De+WL	351.54	18.75	7.20	43.94	7.03	1.35	52.32	49.62	38.26	35.56	25.32	50.97	36.91	45.29	42.59	O.K
112	0.6De+WT	356.54	0.00	129.20	44.57	0.00	24.23	68.79	20.34	68.79	20.34	41.79	44.57	44.57	68.79	20.34	O.K
113	0.6De+EL	336.54	100.00	7.20	42.07	37.50	1.35	80.92	78.22	5.92	3.22	53.92	79.57	4.57	43.42	40.72	O.K
114	0.6De+ET	356.54	0.00	207.20	44.57	0.00	38.85	83.42	5.72	83.42	5.72	56.42	44.57	44.57	83.42	5.72	O.K
115	Do+LL+T	749.04	365.00	70.00	93.63	136.88	13.13	300.00	300.00	0.00	0.00	273.00	300.00	0.00	150.00	150.00	O.K
116	Do+0.75WL+0.75LL	735.29	14.06	70.00	91.91	5.27	13.13	110.31	84.06	99.76	73.51	83.31	97.19	86.64	105.04	78.79	O.K
117	Do+0.75WT+0.75LL	731.54	0.00	161.50	91.44	0.00	30.28	121.72	61.16	121.72	61.16	94.72	91.44	91.44	121.72	61.16	O.K

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201	1.4(Do+T)	829.04	511.00	98.00	103.63	191.63	18.38	420.00	420.00	0.00	0.00	393.00	420.00	0.00	210.00	210.00
202	1.4(Dt+T)	829.04	511.00	98.00	103.63	191.63	18.38	420.00	420.00	0.00	0.00	393.00	420.00	0.00	210.00	210.00
203	1.2Do+1.6WL	762.04	30.00	84.00	95.26	11.25	15.75	122.26	90.76	99.76	68.26	95.26	106.51	84.01	111.01	79.51
204	1.2Dt+1.6WL	762.04	30.00	84.00	95.26	11.25	15.75	122.26	90.76	99.76	68.26	95.26	106.51	84.01	111.01	79.51
205	1.2Do+1.6WT	754.04	0.00	279.20	94.26	0.00	52.35	146.61	41.91	146.61	41.91	119.61	94.26	94.26	146.61	41.91
206	1.2Dt+1.6WT	754.04	0.00	279.20	94.26	0.00	52.35	146.61	41.91	146.61	41.91	119.61	94.26	94.26	146.61	41.91
207	1.2Do+EL	774.04	100.00	84.00	96.76	37.50	15.75	150.01	118.51	75.01	43.51	123.01	134.26	59.26	112.51	81.01
208	1.2Dt+EL	774.04	100.00	84.00	96.76	37.50	15.75	150.01	118.51	75.01	43.51	123.01	134.26	59.26	112.51	81.01
209	1.2Do+ET	754.04	0.00	284.00	94.26	0.00	53.25	147.51	41.01	147.51	41.01	120.51	94.26	94.26	147.51	41.01
210	1.2Dt+ET	754.04	0.00	284.00	94.26	0.00	53.25	147.51	41.01	147.51	41.01	120.51	94.26	94.26	147.51	41.01
211	0.9De+1.6WL	374.79	30.00	10.80	46.85	11.25	2.03	60.12	56.07	37.62	33.57	33.12	58.10	35.60	48.87	44.82
212	0.9De+1.6WT	382.79	0.00	210.80	47.85	0.00	39.53	87.37	8.32	87.37	8.32	60.37	47.85	47.85	87.37	8.32
213	0.9De+EL	362.79	100.00	10.80	45.35	37.50	2.03	84.87	80.82	9.87	5.82	57.87	82.85	7.85	47.37	43.32
214	0.9De+ET	382.79	0.00	210.80	47.85	0.00	39.53	87.37	8.32	87.37	8.32	60.37	47.85	47.85	87.37	8.32
215	1.2Do+1.2T+1.6LL	866.04	438.00	84.00	108.26	164.25	15.75	360.00	360.00	0.00	0.00	333.00	360.00	0.00	180.00	180.00
216	1.2Do+1.6WL+0LL	832.04	30.00	84.00	104.01	11.25	15.75	131.01	99.51	108.51	77.01	104.01	115.26	92.76	119.76	88.26
217	1.2Do+1.6WT+1.0LL	824.04	0.00	279.20	103.01	0.00	52.35	155.36	50.66	155.36	50.66	128.36	103.01	103.01	155.36	50.66
	Max :	866	511	284												
	Min :	337	0	7												

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 273.00 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 293.00 kN/m^2

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 393.00 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 413.00 kN/m^2

Sample calculation of values of base pressure:

Load case : Dt+T

Maximum pressure, p_1 (pmax) = 84.88+136.88+13.13= 234.88 kN/m^2

p_2= 84.88+136.88-13.13= 208.63 kN/m^2

p_3= 84.88-(136.88+13.13)= -65.12 kN/m^2

p_4= 84.88-136.88+13.13= -38.87 kN/m^2

Maxm pressure excluding weight of soil (p_{max}-γh)= 300.00-1.5*18= 273.00 kN/m^2

Maximum average Pressure along X, $p_{avx(max)}$ = (300.00+300.00)/2= 221.76 kN/m^2

Minimum average Pressure along X, $p_{avx(minx)}$ = (0.00+0.00)/2= -52.00 kN/m^2

Maximum average Pressure along Y, $p_{avy(max)}$ = (300.00+0.00)/2= 98.01 kN/m^2

Minimum average Pressure along Y, $p_{avy(minx)}$ = (0.00+300.00)/2= 71.76 kN/m^2

Max PressureDL+LL	320.00	**OK**
Max Pressure Including SL/WL	320.00	**OK**

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Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.65+2*0.5*0.526)+(3.1+2*0.5*0.526))*2

= 9.604 m

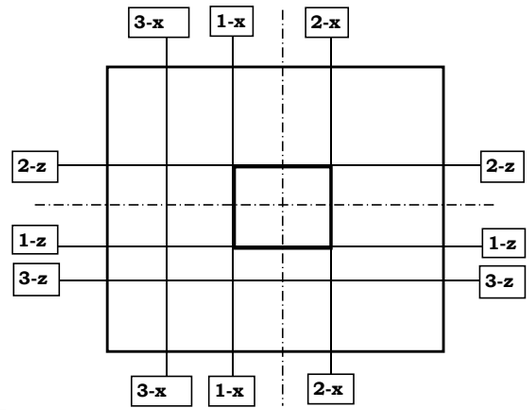
Punching Area= ((0.65+2*0.5*0.526)*(3.1+2*0.5*0.526))

= 4.26 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.6*24+0.9*18)

= 30.60 kN/m^2

Area of footing, A_f= 8.00 m^2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	B.M_{1x} kNm	B.M_{2x} kNm	S.F_{3x} kN	S.$F_{(punching)}$ kN
Ultimate Condition								
201	1.4(Do+T)	278.25	141.75	392.07	77.95	3.79	49.93	273.13
202	1.4(Dt+T)	278.25	141.75	392.07	77.95	3.79	49.93	273.13
203	1.2Do+1.6WL	98.92	91.60	105.01	16.72	12.74	10.00	241.81
204	1.2Dt+1.6WL	98.92	91.60	105.01	16.72	12.74	10.00	241.81
205	1.2Do+1.6WT	94.26	94.26	94.26	14.50	14.50	8.47	238.07
206	1.2Dt+1.6WT	94.26	94.26	94.26	14.50	14.50	8.47	238.07
207	1.2Do+EL	108.95	84.57	129.27	21.69	8.45	13.46	247.42
208	1.2Dt+EL	108.95	84.57	129.27	21.69	8.45	13.46	247.42
209	1.2Do+ET	94.26	94.26	94.26	14.50	14.50	8.47	238.07
210	1.2Dt+ET	94.26	94.26	94.26	14.50	14.50	8.47	238.07
211	0.9De+1.6WL	50.51	43.19	56.60	5.69	1.72	3.56	60.77
212	0.9De+1.6WT	47.85	47.85	47.85	3.93	3.93	2.29	64.51
213	0.9De+EL	57.54	33.16	77.86	9.98	-3.26	6.62	55.16
214	0.9De+ET	47.85	47.85	47.85	3.93	3.93	2.29	64.51
215	1.2Do+1.2T+1.6LL	238.50	121.50	336.06	65.82	2.26	42.22	290.43
216	1.2Do+1.6WL+0LL	107.67	100.35	113.76	18.71	14.74	11.16	274.53
217	1.2Do+1.6WT+1.0LL	103.01	103.01	103.01	16.50	16.50	9.63	270.79

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Sample calculation of BM & SF

Load case : 201

p_{1x} = 420-(420-0)*0.675/2 = 278.25 kN/m^2 Overhang = 0.675 m

p_{2x} = 0+(420-0)*0.675/2 = 141.75 kN/m^2 Effective depth = 0.542 m

p_{3x} = 420-(420-0)*(0.675-0.542)/2 = 392.07 kN/m^2

BM_{1x} = (278.25-30.6)*0.675*0.675/2+0.5*(420-278.25)*0.675*0.675*0.67 = 77.95 kN-m/m

BM_{2x} = (141.75-30.6)*0.675*0.675/2-0.5*(-0+141.75)*0.675*0.675*0.67 = 3.79 kN-m/m

SF_{3x} = (392.07-30.6)*(0.675-0.542)+0.5*(420-392.07)*(0.675-0.542) = 49.93 kN/m

$S.F_{(punching)}$ = (8-4.26)*(103.63-30.6) = 273.13 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	77.95	0.27	0.35	0.001	459.29	975.60	16ϕT	206.09	300	200	1005.31
At top	-3.26	-0.01	0.01	0.000	19.10	975.60	16ϕT	206.09	300	200	1005.31

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, $(\phi\ Vc)/bd$ = 0.740 N/mm2

Shear at a distance 'd' from face of column, Vu= 49.93 kN/m

Shear stress , tv= (49.93*1000)/(1000*542)

= 0.09 N/mm2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 273.13 kN

Strength of Concrete in shear Vc1 = 54089.31 kN Table 22.6.5.2_ (a)

Strength of Concrete in shear Vc2 = 8029.11 kN Table 22.6.5.2_ (b)

Strength of Concrete in shear Vc3 = 10265.12 kN Table 22.6.5.2_ (c)

Design Vc= 8029.11 kN

Allowable shear stress, $(\phi\ Vc)/bd$ = 1.16 N/mm2

Design punching shear stress, τ = (273.13*1000)/(9604*542)

= 0.05 N/mm^2 **No shear reinforcement reqd.**

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Calculation of Bending Moment and Shear Force in Z direction :

p_{1z} =pressure at the face of column at section 1z-1z in kN/m^2.
p_{2z} =pressure at the face of column at section 2z-2z in kN/m^2.
p_{3z} =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
$B.M_{1z}$ =Moment at the face of column at section 1z-1z in kN-m/m.
$B.M_{2z}$ =Moment at the face of column at section 2z-2z in kN-m/m.
$S.F_{3z}$ = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions							
201	1.4(Do+T)	210.00	210.00	210.00	18.16	18.16	-13.63
202	1.4(Dt+T)	210.00	210.00	210.00	18.16	18.16	-13.63
203	1.2Do+1.6WL	107.47	83.05	111.61	8.02	5.07	-6.13
204	1.2Dt+1.6WL	107.47	83.05	111.61	8.02	5.07	-6.13
205	1.2Do+1.6WT	134.83	53.69	148.60	11.35	1.54	-8.89
206	1.2Dt+1.6WT	134.83	53.69	148.60	11.35	1.54	-8.89
207	1.2Do+EL	108.97	84.55	113.11	8.17	5.22	-6.25
208	1.2Dt+EL	108.97	84.55	113.11	8.17	5.22	-6.25
209	1.2Do+ET	135.53	52.99	149.53	11.44	1.45	-8.96
210	1.2Dt+ET	135.53	52.99	149.53	11.44	1.45	-8.96
211	0.9De+1.6WL	48.41	45.28	48.95	1.83	1.45	-1.39
212	0.9De+1.6WT	78.48	17.21	88.87	5.45	-1.96	-4.37
213	0.9De+EL	46.91	43.78	47.45	1.68	1.30	-1.28
214	0.9De+ET	78.48	17.21	88.87	5.45	-1.96	-4.37
215	1.2Do+1.2T+1.6LL	180.00	180.00	180.00	15.13	15.13	-11.35
216	1.2Do+1.6WL+0LL	116.22	91.80	120.36	8.91	5.96	-6.80
217	1.2Do+1.6WT+1.0LL	143.58	62.44	157.35	12.24	2.42	-9.56

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Sample calculation of BM & SF

Load case : 201

p_{1z} = 210-(210-210)*0.45/4= 210.00 kN/m^2

p_{2z} = 210+(210-210)*0.45/4= 210.00 kN/m^2

p_{3z} = 210-(210-210)*(0.45-0.526)/4= 210.00 kN/m^2

Overhang = 0.45 m

Effective depth = 0.526 m

BM_{1z} = (210.00-30.6)*0.45*0.45/2+0.5*(210-210.00)*0.45*0.45*0.67 = 18.16 kN-m/m

BM_{2z} = (210.00-30.6)*0.45*0.45/2-0.5*(-210+210.00)*0.45*0.45*0.67 = 18.16 kN-m/m

SF_{3z} = (210.00-30.6)*(0.45-0.526)+0.5*(210-210.00)*(0.45-0.526) = -13.63 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	18.16	0.066	0.088	0.0002	113.13	946.80	16ϕT	212.36	300	200	1005.3
At top	-1.96	-0.007	-0.009	0.0000	-0.12	946.80	16ϕT	212.36	300	200	1005.3

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm2

Shear at a distance 'd' from face of column, V_u= -6.13 kN/m

Shear stress , τ_v= (-6.13*1000)/(1000*526)

= -0.01 N/mm2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				
	At Bottom	16ϕT	200	c/c
	At Top	16ϕT	200	c/c
Along Z Direction :				
	At Bottom	16ϕT	200	c/c
	At Top	16ϕT	200	c/c

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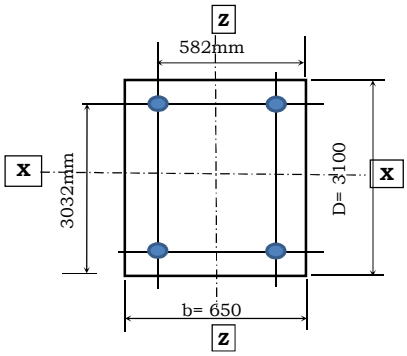
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8.4 Design calculation of Column of 49FE-TNK-006

COLUMN MKD C1

Input:

Clear cover to reinf.=	50mm	Table 20.6.1.3.1
Dia. of main reinf.=	16mm	
Dia. of link=	10mm	
Effective cover =	68mm	
Grade of Concrete (f_c') =	M35	
Grade of Steel (f_y) =	Fy420	
Grade of Reinf(links).=	Fy420	
Min. Steel Ratio	0.15%	
Max. Steel Ratio	4.00%	
Self weight of Column =	87.05kN	



For Compression-controlled sections

Strength reduction factor ϕ= 0.65 Table 21.2.2

Unsupported length of column about X (l_{ux}) =	1800mm
Unsupported length of column about Z (l_{uz}) =	1800mm

For Columns not braced against sideway, Where, Effective length factor K= 2

Slenderness ratio,	$K\,l_{ux} / r_x$ =	3.87	< 22 SECTION 6.2.5a	Radius of gyration r_x=	930	mm
	$K\,l_{uz} / r_z$ =	18.46	< 22 SECTION 6.2.5a	Radius of gyration r_z=	195	mm

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γ_y = 0.96

γ_z = 0.82

Load Case	Load Combination	Factored Load From STAAD			Load at Column Base			Nominal Load/Moment			Kn = Pn/f_cbD (N/mm^2)
		Pu (kN)	Hux (kN)	Huz (kN)	Pu (kN)	Muz (kNm)	Mux (kNm)	Pn (kN)	Mnz (kNm)	Mnx (kNm)	
201	1.4(Do+T)	525.00	204.4	0.0	612.05	367.9	0.0	941.6	566.0	0.0	0.013
202	1.4(Dt+T)	525.00	204.4	0.0	612.05	367.9	0.0	941.6	566.0	0.0	0.013
203	1.2Do+1.6WL	458.00	12.0	0.0	545.05	21.6	0.0	838.5	33.2	0.0	0.012
204	1.2Dt+1.6WL	458.00	12.0	0.0	545.05	21.6	0.0	838.5	33.2	0.0	0.012
205	1.2Do+1.6WT	450.00	0.0	35.2	537.05	0.0	63.4	826.2	0.0	97.5	0.012
206	1.2Dt+1.6WT	450.00	0.0	35.2	537.05	0.0	63.4	826.2	0.0	97.5	0.012
207	1.2Do+EL	470.00	40.0	0.0	557.05	72.0	0.0	857.0	110.8	0.0	0.012
208	1.2Dt+EL	470.00	40.0	0.0	557.05	72.0	0.0	857.0	110.8	0.0	0.012
209	1.2Do+ET	450.00	0.0	40.0	537.05	0.0	72.0	826.2	0.0	110.8	0.012
210	1.2Dt+ET	450.00	0.0	40.0	537.05	0.0	72.0	826.2	0.0	110.8	0.012
211	0.9De+1.6WL	70.75	12.0	0.0	157.80	21.6	0.0	242.8	33.2	0.0	0.003
212	0.9De+1.6WT	78.75	0.0	40.0	165.80	0.0	72.0	255.1	0.0	110.8	0.004
213	0.9De+EL	58.75	40.0	0.0	145.80	72.0	0.0	224.3	110.8	0.0	0.003
214	0.9De+ET	78.75	0.0	40.0	165.80	0.0	72.0	255.1	0.0	110.8	0.004
215	1.2Do+1.2T+1.6LL	562.00	175.2	0.0	649.05	315.4	0.0	998.5	485.2	0.0	0.014
216	1.2Do+1.6WL+0LL	528.00	12.0	0.0	615.05	21.6	0.0	946.2	33.2	0.0	0.013
217	1.2Do+1.6WT+1.0LL	520.00	0.0	35.2	607.05	0.0	63.4	933.9	0.0	97.5	0.013

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Reinforcement: Provide,

4-Nos	16 mm Dia
22-Nos	16 mm Dia
4-Nos	16 mm Dia

804.57 Sq mm (At Corner)
4425.14 Sq mm (Along Z axis)
804.57 Sq mm (Along Y axis)

A_{sc} pvd. = 6034.29 Sq mm

Total A_{sc} = 0.30 % ρ_g = 0.003

β = P_n / P_0 (range is 0.55 to 0.70) Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 6230.113 T SECTION 22.4.2.2

Load Comb	From Chart of SP17(M)-R35-420.8		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f_cb^2D$	$R_{nx}=M_{nox}/f_cb^2D$						
1.4(Do+T)	0.050	0.050	2292.1	10931.4	0.55	1.16	0.20	O.K
1.4(Dt+T)	0.050	1.050	2292.1	229558.9	0.55	1.16	0.20	O.K
1.2Do+1.6WL	0.050	2.050	2292.1	448186.4	0.55	1.16	0.01	O.K
1.2Dt+1.6WL	0.050	3.050	2292.1	666813.9	0.55	1.16	0.01	O.K
1.2Do+1.6WT	0.050	4.050	2292.1	885441.4	0.55	1.16	0.00	O.K
1.2Dt+1.6WT	0.050	5.050	2292.1	1104068.9	0.55	1.16	0.00	O.K
1.2Do+EL	0.050	6.050	2292.1	1322696.4	0.55	1.16	0.03	O.K
1.2Dt+EL	0.050	7.050	2292.1	1541323.9	0.55	1.16	0.03	O.K
1.2Do+ET	0.050	8.050	2292.1	1759951.4	0.55	1.16	0.00	O.K
1.2Dt+ET	0.050	9.050	2292.1	1978578.9	0.55	1.16	0.00	O.K
0.9De+1.6WL	0.050	10.050	2292.1	2197206.4	0.55	1.16	0.01	O.K
0.9De+1.6WT	0.050	11.050	2292.1	2415833.9	0.55	1.16	0.00	O.K
0.9De+EL	0.050	12.050	2292.1	2634461.4	0.55	1.16	0.03	O.K
0.9De+ET	0.050	13.050	2292.1	2853088.9	0.55	1.16	0.00	O.K
1.2Do+1.2T+1.6LL	0.050	14.050	2292.1	3071716.4	0.55	1.16	0.17	O.K
1.2Do+1.6WL+0LL	0.050	15.050	2292.1	3290343.9	0.55	1.16	0.01	O.K
1.2Do+1.6WT+1.0LL	0.050	16.050	2292.1	3508971.4	0.55	1.16	0.00	O.K

Links, 10 mm dia @ 250 C/C

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8.5 Design calculation of Footing of 49FE-TNK-009

Size of Columns at node	Along-X	Along-Z
1	0.35m	1.60m
3	0.35m	1.60m
Size of fdn. raft:	2.00m	2.00m
Thickness of raft:	0.50	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.60	m
U/S level of fdn:	-1.50	m
Unit weight of conc:	25	kN/m^3
Unit weight of soil:	18	kN/m^3

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm

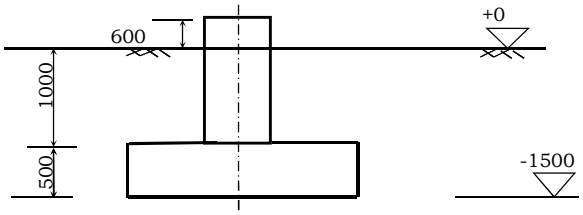
Calculation of Weight of Foundation :

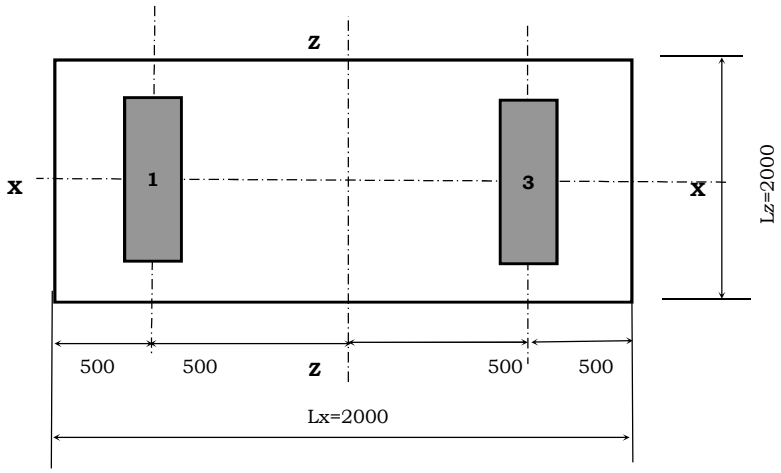
Weight of raft = (2*2*0.5*25) = 50.00KN

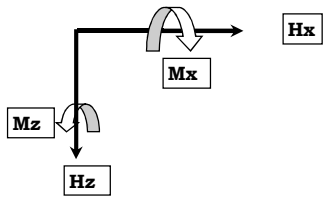
Weight of soil = ((2*2)-2*(0.35*1.6))*1*18 = 51.84KN

Total = 101.84KN

Submerged wt. = 101.84-2*2*1.5*10 = 41.84KN







H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing

Load at bottom of fooring	Node 1					Node 14				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
101 DO + T	41.7	6.0	-12.6	0.0	-1.5	41.7	-6.0	12.6	0.0	-1.5
102 DT + T	41.7	6.0	-12.6	0.0	-1.5	41.7	-6.0	12.6	0.0	-1.5
103 DO + WL	45.3	1.0	-2.1	0.0	-1.5	38.1	1.0	-2.1	0.0	-1.5
104 DT + WL	45.3	1.0	-2.1	0.0	-1.5	38.1	1.0	-2.1	0.0	-1.5
105 DO + WT	41.7	0.0	0.0	-0.8	-4.9	41.7	0.0	0.0	-0.8	-4.9
106 DT + WT	41.7	0.0	0.0	-0.8	-4.9	41.7	0.0	0.0	-0.8	-4.9
107 DO + 0.7 ELo	41.7	1.5	-3.2	0.0	-1.5	41.7	1.5	-3.2	0.0	-1.5
108 DT + 0.7 ELo	41.7	1.5	-3.2	0.0	-1.5	41.7	1.5	-3.2	0.0	-1.5
109 DO + 0.7 ETo	41.7	0.0	0.0	-1.5	-4.7	41.7	0.0	0.0	-1.5	-4.7
110 DT + 0.7 ETo	41.7	0.0	0.0	-1.5	-4.7	41.7	0.0	0.0	-1.5	-4.7
111 0.6DE + WL	25.0	1.0	-2.1	0.0	-0.6	17.8	1.0	-2.1	0.0	-0.6
112 0.6DE + WT	21.4	0.0	0.0	-0.8	-4.0	21.4	0.0	0.0	-0.8	-4.0
113 0.6DE+ELe	21.4	1.0	-2.1	0.0	-0.6	21.4	1.0	-2.1	0.0	-0.6
114 0.6DE + ETe	21.4	0.0	0.0	-1.0	-2.7	21.4	0.0	0.0	-1.0	-2.7
115 DO + LL + T	41.7	6.0	-12.6	0.0	-1.5	41.7	-6.0	12.6	0.0	-1.5
116 DO + 0.75WL + 0.75LL	44.4	0.8	-1.6	0.0	-1.5	39.0	0.8	-1.6	0.0	-1.5
117 DO + 0.75WT + 0.75LL	41.7	0.0	0.0	-0.6	-4.0	41.7	0.0	0.0	-0.6	-4.0

hj,

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Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing								
	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
101 DO + T	185.3	0.00	0.0	0.0	-3.0	NA	61.8	NA	NA
102 DT + T	185.3	0.00	0.0	0.0	-3.0	NA	61.8	NA	NA
103 DO + WL	185.3	2.00	-7.8	0.0	-3.0	23.8	61.8	37.1	NA
104 DT + WL	185.3	2.00	-7.8	0.0	-3.0	23.8	61.8	37.1	NA
105 DO + WT	185.3	0.00	0.0	-1.7	-9.8	NA	19.0	NA	44.9
106 DT + WT	185.3	0.00	0.0	-1.7	-9.8	NA	19.0	NA	44.9
107 DO + 0.7 ELo	185.3	3.00	-6.3	0.0	-3.0	29.4	61.8	24.7	NA
108 DT + 0.7 ELo	185.3	3.00	-6.3	0.0	-3.0	29.4	61.8	24.7	NA
109 DO + 0.7 ETo	185.3	0.00	0.0	-3.0	-9.3	NA	19.9	NA	24.7
110 DT + 0.7 ETo	185.3	0.00	0.0	-3.0	-9.3	NA	19.9	NA	24.7
111 0.6DE + WL	144.7	2.00	-7.8	0.0	-1.2	18.6	120.6	28.9	NA
112 0.6DE + WT	144.7	0.00	0.0	-1.7	-8.0	NA	18.2	NA	35.1
113 0.6DE+ELe	144.7	2.00	-4.2	0.0	-1.2	34.5	120.6	28.9	NA
114 0.6DE + ETe	144.7	0.00	0.0	-2.0	-5.4	NA	26.8	NA	28.9
115 DO + LL + T	185.3	0.00	0.0	0.0	-3.0	NA	61.8	NA	NA
					Min.=	**18.6**	**18.2**	**24.7**	**24.7**

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Base pressure check of footing

	Node	L/C	Horizontal Fx N/mm2	Vertical Fy N/mm2	Horizontal Fz N/mm2
Max Px	1	101	0	0.05	0
Min Px	1	101	0	0.05	0
Max Py	5	103	0	0.06	0
Min Py	8	111	0	0.02	0
Max Pz	1	101	0	0.05	0
Min Pz	1	101	0	0.05	0

Gross Maximum Base pressure, p_{max} = 58.0 KN/m^2

Net Allowable Safe Bearing Capacity = 450 KN/m^2
Depth of soil above footing = 0.8 m
Unit weight of soil = 18 KN/m^3

Gross Safe Bearing Capacity = 464.4 KN/m^2
> 58.0 KN/m^2
Hence OK

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Design of foundation raft F8 & F9

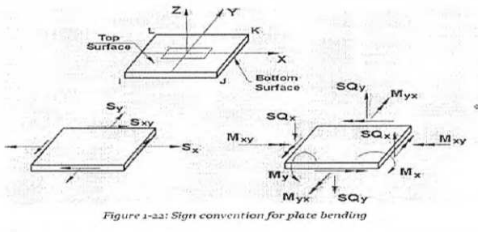
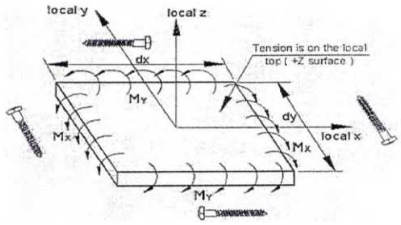


Figure 1-2a: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F8&F9_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	208	0.263	-0.216	0	0	0	31.864	17.69	-19.34	51.204	37.03	101 DO + T
Min Qx	219	-0.263	-0.216	0	0	0	31.864	17.69	0	31.864	17.69	101 DO + T
Max Qy	188	0.253	0.19	0	0	0	31.413	10.934	15.209	46.622	26.143	101 DO + T
Min Qy	208	0.263	-0.216	0	0	0	31.864	17.69	-19.34	51.204	37.03	101 DO + T
Max Sx	4	-0.002	0.004	0	0	0	-0.039	0.049	0.017	-0.056	0.066	101 DO + T
Min Sx	4	-0.002	0.004	0	0	0	-0.039	0.049	0.017	-0.056	0.066	101 DO + T
Max Sy	4	-0.002	0.004	0	0	0	-0.039	0.049	0.017	-0.056	0.066	101 DO + T
Min Sy	4	-0.002	0.004	0	0	0	-0.039	0.049	0.017	-0.056	0.066	101 DO + T
Max Sxy	4	-0.002	0.004	0	0	0	-0.039	0.049	0.017	-0.056	0.066	101 DO + T
Min Sxy	4	-0.002	0.004	0	0	0	-0.039	0.049	0.017	-0.056	0.066	101 DO + T
Max Mx	208	0.263	-0.216	0	0	0	31.864	17.69	-19.34	51.204	37.03	101 DO + T
Min Mx	189	-0.057	0.106	0	0	0	-25.61	6.766	19.927	-45.532	26.693	101 DO + T
Max My	209	-0.176	-0.204	0	0	0	7.503	23.215	6.484	13.987	29.699	105 DO + WT
Min My	188	0.068	0.048	0	0	0	2.901	-2.697	-5.642	8.543	-8.339	112 0.6DE + WT
Max Mxy	189	-0.057	0.106	0	0	0	-25.61	6.766	19.927	-45.532	26.693	101 DO + T
Min Mxy	198	0.057	0.106	0	0	0	-25.61	6.766	-19.927	-45.532	26.693	101 DO + T

PLATE CENTER STRESS : FDN_F8&F9_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	208	0.263	-0.216	0	0	0	31.864	17.69	-19.34	51.204	37.03	101 DO + T
Min Qx	219	-0.263	-0.216	0	0	0	31.864	17.69	0	31.864	17.69	101 DO + T
Max Qy	188	0.253	0.19	0	0	0	31.413	10.934	15.209	46.622	26.143	101 DO + T
Min Qy	208	0.263	-0.216	0	0	0	31.864	17.69	-19.34	51.204	37.03	101 DO + T
Max Sx	4	-0.002	0.004	0	0	0	-0.039	0.049	0.017	-0.056	0.066	101 DO + T
Min Sx	4	-0.002	0.004	0	0	0	-0.039	0.049	0.017	-0.056	0.066	101 DO + T
Max Sy	4	-0.002	0.004	0	0	0	-0.039	0.049	0.017	-0.056	0.066	101 DO + T
Min Sy	4	-0.002	0.004	0	0	0	-0.039	0.049	0.017	-0.056	0.066	101 DO + T
Max Sxy	4	-0.002	0.004	0	0	0	-0.039	0.049	0.017	-0.056	0.066	101 DO + T
Min Sxy	4	-0.002	0.004	0	0	0	-0.039	0.049	0.017	-0.056	0.066	101 DO + T
Max Mx	208	0.263	-0.216	0	0	0	31.864	17.69	-19.34	51.204	37.03	101 DO + T
Min Mx	189	-0.057	0.106	0	0	0	-25.61	6.766	19.927	-45.532	26.693	101 DO + T
Max My	209	-0.176	-0.204	0	0	0	7.503	23.215	6.484	13.987	29.699	105 DO + WT
Min My	188	0.068	0.048	0	0	0	2.901	-2.697	-5.642	8.543	-8.339	112 0.6DE + WT
Max Mxy	189	-0.057	0.106	0	0	0	-25.61	6.766	19.927	-45.532	26.693	101 DO + T
Min Mxy	198	0.057	0.106	0	0	0	-25.61	6.766	-19.927	-45.532	26.693	101 DO + T

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

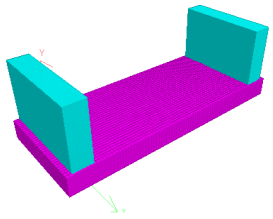
Force Values			Depth		Reinforcement Calculation													
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
101 DO +	51.20	top face	50	444	51.204	0.35	0.0008	368.3	399.6	12	283.14	200	565.8	7.978	9.972	0.131	tension control
101 DO +	-45.53	bottom face	50	444	45.532	0.31	0.0007	327.3		12	283.14	200	565.8	7.978	9.972	0.131	tension control
101 DO +	-45.53	top face	50	444	45.532	0.31	0.0007	327.3		12	283.14	200	565.8	7.978	9.972	0.131	tension control
101 DO +	-45.53	top face	50	444	45.532	0.31	0.0007	327.3		12	283.14	200	565.8	7.978	9.972	0.131	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Rema rks	φV_c	Remar ks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN			
Load condition	Max Qx	0.263	1	131.5	51.204	0.001	0.444	431.246	429.897	761.754	429.9	1623	> 131.5	ok	322.4	φVc>Vu	not required
	Min Qx	0.263	1	131.5	31.864			437.903			429.9	1623	> 131.5	ok	322.4	φVc>Vu	not required

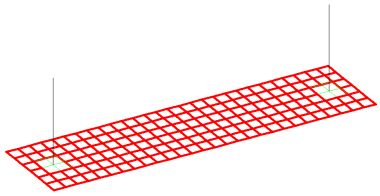
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation													
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
105 DO +	29.70	top face	50	444	29.699	0.20	0.0005	213.1	399.6	12	283.14	200	565.8	7.978	9.972	0.131	tension control
112 0.6D	-8.34	bottom face	50	444	8.339	0.06	0.0001	59.7		12	283.14	200	565.8	7.978	9.972	0.131	tension control
101 DO +	26.69	bottom face	50	444	26.693	0.18	0.0004	191.4		12	283.14	200	565.8	7.978	9.972	0.131	tension control
101 DO +	26.69	top face	50	444	26.693	0.18	0.0004	191.4		12	283.14	200	565.8	7.978	9.972	0.131	tension control

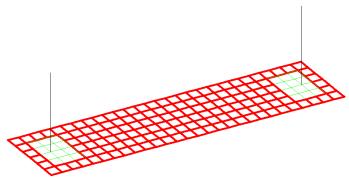
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Rema rks	φV_c	Remar ks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN			
Load condition	101 DO +	0.19	1	95	26.143	0.001	0.444	435.797	429.897	fc'761.8	429.9	452	> 95	ok	322.4	φVc>Vu	not required
	101 DO +	0.216	1	108	37.03			432.734			429.9	452	> 108	ok	322.4	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	12ϕT	200	c/c
At Top	12ϕT	200	c/c
Along Z Direction :			
At Bottom	12ϕT	200	c/c
At Top	12ϕT	200	c/c

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REFERENCES:

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Document. No. MD-216-1000-EG-ST- CRT-1001) Clause 4.1.

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ATTACHMENT 1 STAAD INPUT FILE FOR 49FE-TNK-009:

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STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 03-Dec-20
END JOB INFORMATION
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UNIT METER KN
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DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007

POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY
1 2 PRIS YD 0.35 ZD 1.6
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
4 TO 403 PLATE MAT DIRECT Y SUBGRADE 40500 PRINT
LOAD 1 LOADTYPE None TITLE EMPTY LOAD (DE)
JOINT LOAD
2 4 FY -8
2 4 MX 1
SELFWEIGHT Y -1 LIST ALL
ELEMENT LOAD
4 TO 403 PR GY -18
LOAD 2 LOADTYPE None TITLE OPEARTING LOAD, DO
JOINT LOAD
2 4 FY -14
2 4 MX 1.5
SELFWEIGHT Y -1 LIST ALL
ELEMENT LOAD
4 TO 403 PR GY -18
LOAD 3 LOADTYPE None TITLE TEST LOAD, DT
JOINT LOAD
2 4 FY -14 MX 1.5
SELFWEIGHT Y -1 LIST ALL
ELEMENT LOAD
4 TO 403 PR GY -18
LOAD 4 LOADTYPE None TITLE TEMPARATURE LOAD, T
JOINT LOAD
4 FX 6
2 FX -6
LOAD 5 LOADTYPE None TITLE WIND LONG, WL
JOINT LOAD
2 FX -1 FY -3.6
4 FX -1 FY 3.6
LOAD 6 LOADTYPE None TITLE WIND TRANSVERSE, WT
JOINT LOAD
2 4 FZ 0.825 MX 1.65
LOAD 7 LOADTYPE None TITLE Empty Condition SEISMIC LONG, ELe
JOINT LOAD

2 4 FX -1
LOAD 8 LOADTYPE None TITLE Empty Condition SEISMIC TRANS, ETe
JOINT LOAD
2 4 FZ 1
LOAD 9 LOADTYPE None TITLE LIVE LOAD, LL
LOAD 10 LOADTYPE None TITLE Operating Condition SEISMIC LONG, ELo
JOINT LOAD
2 4 FX -1.5
LOAD 11 LOADTYPE None TITLE Operating Condition SEISMIC TRANS, ETo
JOINT LOAD
2 4 FZ 1.5
LOAD COMB 101 DO + T
2 1.0 4 1.0
LOAD COMB 102 DT + T
3 1.0 4 1.0
LOAD COMB 103 DO + WL
2 1.0 5 1.0
LOAD COMB 104 DT + WL
3 1.0 5 1.0
LOAD COMB 105 DO + WT
2 1.0 6 1.0
LOAD COMB 106 DT + WT
3 1.0 6 1.0
LOAD COMB 107 DO + 0.7 ELo
2 1.0 10 1.0
LOAD COMB 108 DT + 0.7 ELo
3 1.0 10 1.0
LOAD COMB 109 DO + 0.7 ETo
2 1.0 11 1.0
LOAD COMB 110 DT + 0.7 ETo
3 1.0 11 1.0
LOAD COMB 111 0.6DE + WL
1 0.6 5 1.0
LOAD COMB 112 0.6DE + WT
1 0.6 6 1.0
LOAD COMB 113 0.6DE+ELe
1 0.6 7 1.0
LOAD COMB 114 0.6DE + ETe
1 0.6 8 1.0
LOAD COMB 115 DO + LL + T
2 1.0 9 1.0 4 1.0
LOAD COMB 116 DO + 0.75WL + 0.75LL
2 1.0 9 0.75 5 0.75
LOAD COMB 117 DO + 0.75WT + 0.75LL
2 1.0 6 0.75 9 0.75
LOAD COMB 201 1.4 (DO + T)

2 1.4 4 1.4
LOAD COMB 202 1.4 (DT + T)
3 1.4 4 1.4
LOAD COMB 203 1.2DO + 1.6WL
2 1.2 5 1.6
LOAD COMB 204 1.2 DT + 1.6WL
3 1.2 5 1.6
LOAD COMB 205 1.2DO + 1.6WT
2 1.2 6 1.6
LOAD COMB 206 1.2DT + 1.6WT
3 1.2 6 1.6
LOAD COMB 207 1.2DO + ELo
2 1.2 10 1.2
LOAD COMB 208 1.2DT + ELo
3 1.2 10 1.2
LOAD COMB 209 1.2DO+ETo
2 1.2 11 1.2
LOAD COMB 210 1.2DT+ETo
3 1.2 11 1.0
LOAD COMB 211 0.9DE+1.6WL
1 0.9 5 1.6
LOAD COMB 212 0.9DE+1.6WT
1 0.9 6 1.6
LOAD COMB 213 0.9DE+ELe
1 0.9 7 1.0
LOAD COMB 214 0.9DE+ETe
1 0.9 8 1.0
LOAD COMB 215 1.2DO+1.2T+1.6LL
2 1.2 4 1.2 9 1.6
LOAD COMB 216 1.2DO+1.6WL+1.0LL
2 1.2 5 1.6 9 1.0
LOAD COMB 217 1.2DO+1.6WT+1.0LL
2 1.2 6 1.6 9 1.0
PERFORM ANALYSIS
FINISH

ATTACHMENT 2 EQUIPMENT DRAWING

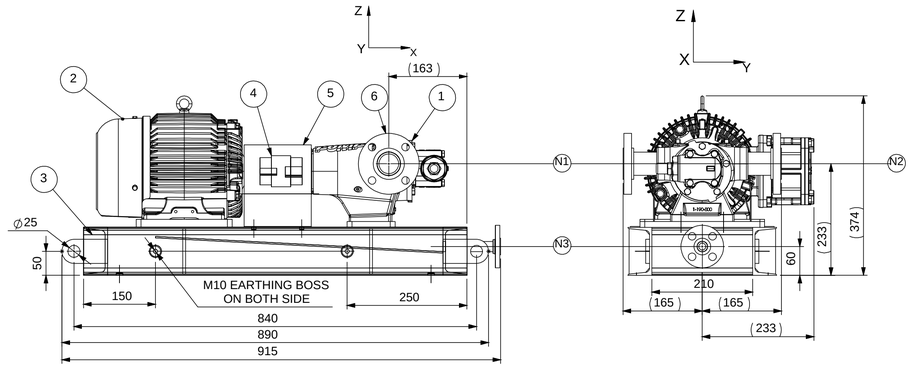
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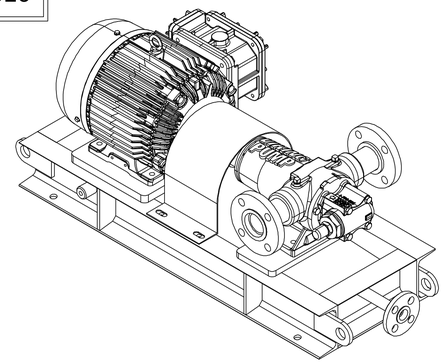
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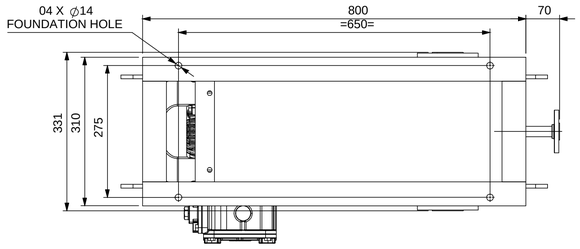
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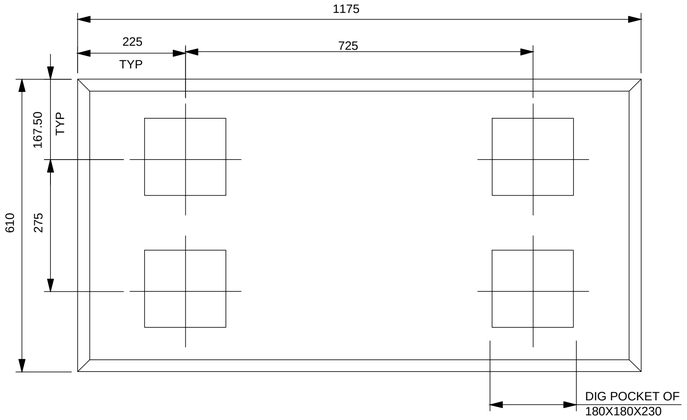
DRAWING NUMBER: MD-216-49FE-EG-ME-GAD-V558-1012 | REV: C01

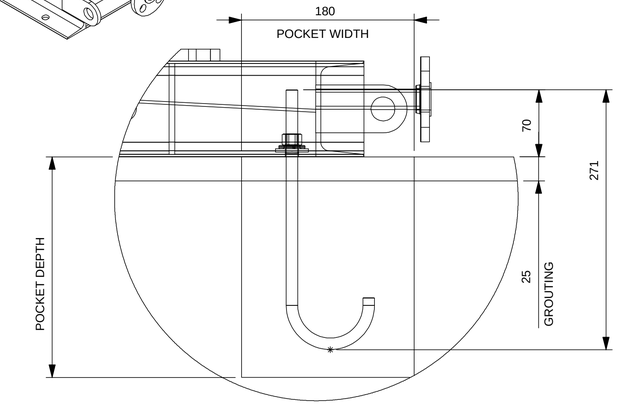
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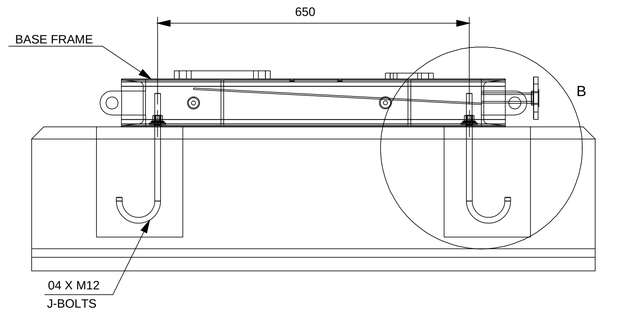








DETAIL B
SCALE 2 : 13



NOTES:
1) ALL DIMENSIONS ARE IN mm.
2) ALL DIMENSIONS TOLERANCE ±5mm.
3) FOUNDATION BOLTS SHALL BE SUPPLIED BY IDEX.

PUMP PERFORMANCE DATA			
PUMP	GD^2 (kg-m^2)	FLOWRATE (LPM)	MAX. DIFF. PR. [bar(g)]
HL 4195	0.0058	95	2.9

MARK	QTY.	SIZE	CLASS	SERVICE
N1	1	1.5" SORF	#150	SUCTION
N2	1	1.5" SORF	#150	DISCHARGE
N3	1	0.5" SORF	#150	DRAIN

ALLOWABLE NOZZLE LOADING (FORCES & MOMENTS)								
NOZZLE NO	NOZZLE	SIZE	FORCE IN NEWTON			MOMENTS IN NEWTON - METER		
			PX	PY	PZ	MX	MY	MZ
N1	SUCTION	1.5"	500	500	500	250	250	35
N2	DISCHARGE	1.5"	500	500	500	250	250	35

ITEM NO.	PART NUMBER	DESCRIPTION C01	MATERIAL / MAKE	QTY./ SET
1	PUMP	MODEL - HL 4195 TAG NUMBER : 49FE-PMP-016 MAKE : VIGEN	SHALL BE AS PER PUMP DATA SHEET (MA'ADEN DOC. NO. MD-216-49FE-EG-ME-DAT- V558-1001) MAKE : VIGEN	1
2	MOTOR	2.2 kW, 4 POLE FOOT MOUNTED TAG: 49FE-PMP-016-MTR	SHALL BE AS PER MOTOR DATA SHEET (MA'ADEN DOC. NO. MD-216-49FE-EG-EL-DAT- V558-1001)	1
3	BASE FRAME	ISMC 100 X 50	GR. E250 BR / B0	1
4	COUPLING	STD	FENNER	1
5	COUPLING GUARD	PAINTED	CRCA	1

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		C01	17/10/2020	ISSUED FOR CONSTRUCTION AND REVISED AS PER EMPLOYER COMMENTS DATED 13/09/2020	--	SRM	--	KBJ	
MD-216-49FE-EG-PR-PID-V558-1009	FUEL OIL STORAGE PID FOR PETROL	A01	11/09/2020	ISSUED FOR REFERENCE	--	SRM	--	KBJ	
	UNLOADING STORAGE & DISTRIBUTION								

ABUNAYYAN-UNIGAZ

DRAWING NUMBER					REVISION
2006-UNI-DWG-055					1
DATE	08.09.2020	DATE	09.09.2020	DATE	10.09.2020
DRAWN	JS	CHECKED	PKS	APPROVED	SR

LARSEN & TOUBRO DRAWING NUMBER: 019045_080_U_V_19146_006 — REVISION: 1

OUTOTEC DRAWING NUMBER: 900707 _GAU01 _AG50_MTL01 _49022 — DOCUMENT ID: 0U100694272 — REVISION: 1

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Saudi Arabian Mining Company شركة معادن العربية السعودية

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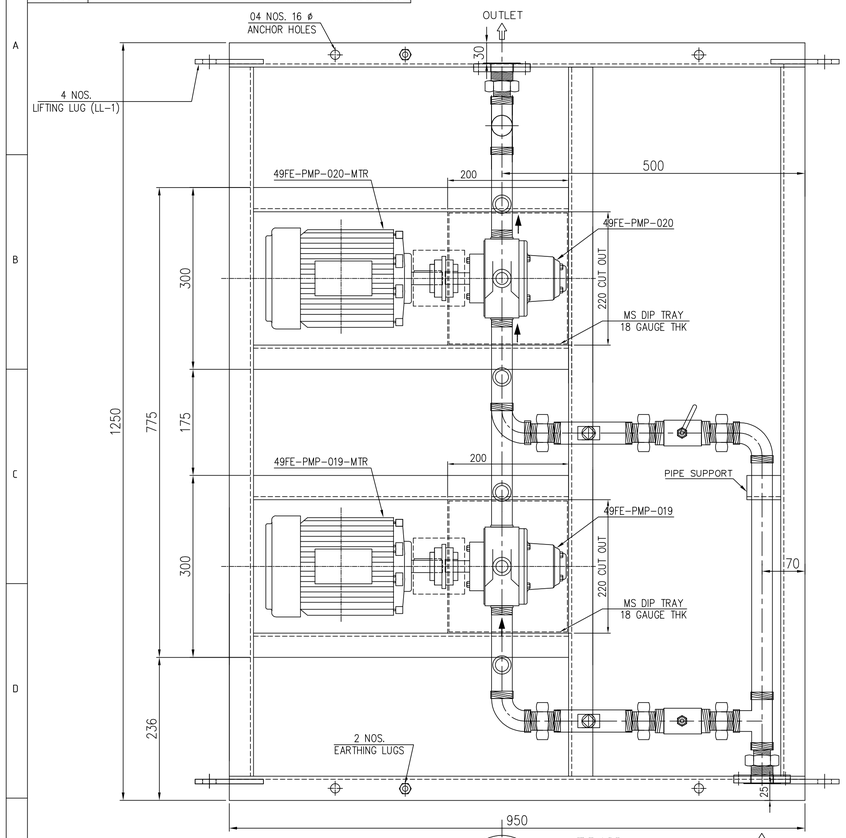
DRAWN	--	--	
CHECKED	SRM	10/09/2020	
ENGINEER	--	--	
APPROVED	KBJ	11/09/2020	
EMPLOYER			
SCALE	NTS		A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
FUEL OIL STORAGE
DRAIN PUMP FOR PETROL DISPENSING AREA

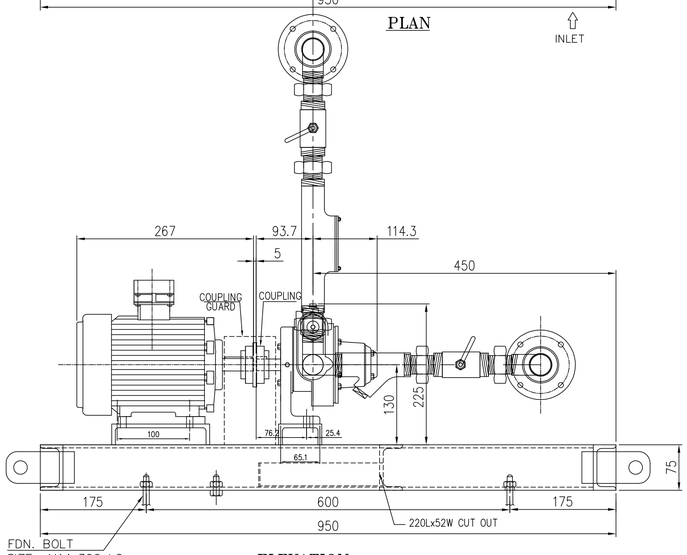
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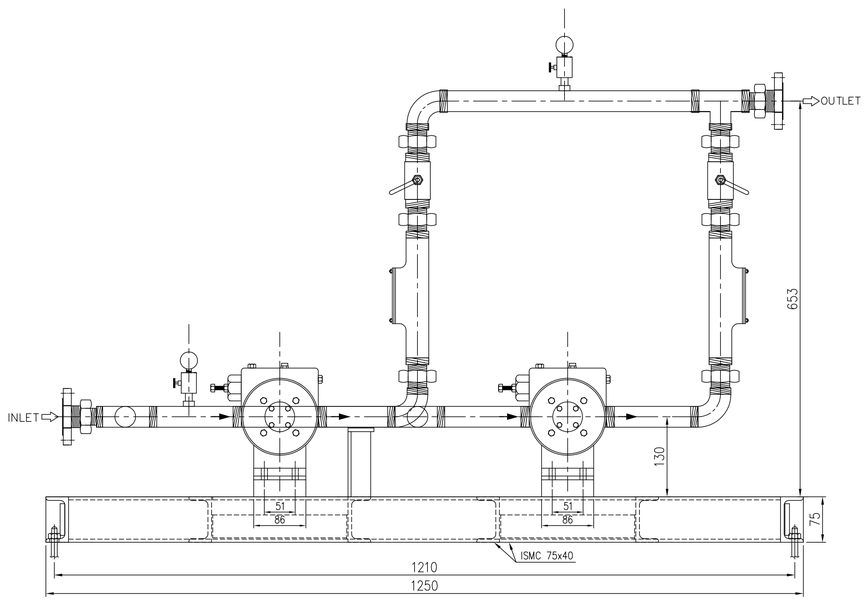
TAG NO: 49FE-PMP-019
49FE-PMP-020



PLAN



ELEVATION



SIDE VIEW

S.NO.	ITEMS	SIZE (A02)	QTY. (1 NO. SKID)
01	PUMP (TAG NO: 49FE-PMP-019 & 49FE-PMP-020)	XB1PC	02 NOS.
02	MOTOR (TAG NO: 49FE-PMP-019-MTR & 49FE-PMP-020-MTR)	0.55 kW	02 NOS.
03	SHAFT COUPLING	MAKE : FENNER SIZE/MODEL : F110 TYPE : LOVE JOY COUPLING	02 NOS.
04	THREADED BALL VALVE (INLET SIDE)	25NB, ENDS-NPT(F)	02 NOS.
05	THREADED STRAINER (INLET SIDE)	25NB, ENDS-NPT(F)	02 NOS.
06	THREADED NON RETURN VALVE (OUTLET SIDE)	25NB, ENDS-NPT(F)	02 NOS.
07	THREADED BALL VALVE (OUTLET SIDE)	25NB, ENDS-NPT(F)	02 NOS.
08	PRESSURE GAUGE (INLET SIDE)	15NB, ENDS-NPT(M) (A02)	01 NO.
09	PRESSURE GAUGE (OUTLET SIDE)	15NB, ENDS-NPT(M)	01 NO.
10	THREADED FLANGE	50NBx150#, ENDS-NPT(F)	02 NOS.
11	THREADED HEX NIPPLE	25NBx25NB, ENDS-NPT(M)	12 NOS.
12	THREADED HEX NIPPLE	25NBx50NB, ENDS-NPT(M)	02 NOS.
13	THREADED ELBOW	25NB, ENDS-NPT(F)	02 NOS.
14	THREADED TEE	25NBx25NB, ENDS-NPT(F)	02 NOS.
15	PIPE LINE	25NBxSCH40	02 Mtr.
16	BSAE FRAME	ISMC 75x40	8 Mtr.
17	LIFTING LUG	8 MM THK.	04 NOS.
18	EARTHING LUGS	--	02 NOS.

VANE PUMP DATA SHEET (A02)

PUMP MAKE	: BLACKMER
PUMP MODEL	: XB1PC
PUMP FLOW RATE	: 27.64 LPM
DIFF. HEAD/PR.RATED	: 4.0 BAR
SERVICE	: JET FUEL
MOTOR TYPE	: FOOT MOUNTED
MOTOR RPM/Hz	: 1110/60 ±5%
MOTOR RATING	: 0.55 kW/60±5% Hz/6 POLE
MOTOR LOAD GD^2	: 0.01 kg-m^2
MOTOR CERTIFICATION	: ATEX APPROVED
QTY. OF SKID	: 01 NOS.
WEIGHT OF 1 NO. SKID	: 250 kg (APPROX.)

NOTES:-
1) ALL DIMENSION ARE IN MM. UNLESS OTHERWISE SPECIFIED.

FDN. BOLT
SIZE:-M14x300 LG.
POCKET-150x150x300mmDEEP

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-49FE-EG-PR-PID-V558-1005	FUEL OIL STORAGE PID FOR JET FUEL UNLOADING,	A02	02/11/2020	REVISED AS PER EMPLOYER COMMENTS DATED 20/10/2020	--	SRM	--	KBJ	
	STORAGE & DISTRIBUTION	A01	11/09/2020	ISSUED FOR REFERENCE	--	SRM	--	KBJ	

ABUNAYYAN-UNIGAZ

DRAWING NUMBER					REVISION
2006-UNI-DWG-057					0
DATE	07.09.2020	DATE	08.09.2020	DATE	09.09.2020
DRAWN	JS	CHECKED	PKS	APPROVED	SR

LARSEN & TOUBRO DRAWING NUMBER: 019045_080_U_V_19146_008 — REVISION: 1

OUTOTEC DRAWING NUMBER: 900707__GAU01__AG50__MTL01__49024 — DOCUMENT ID: OU100694274 — REVISION: 1

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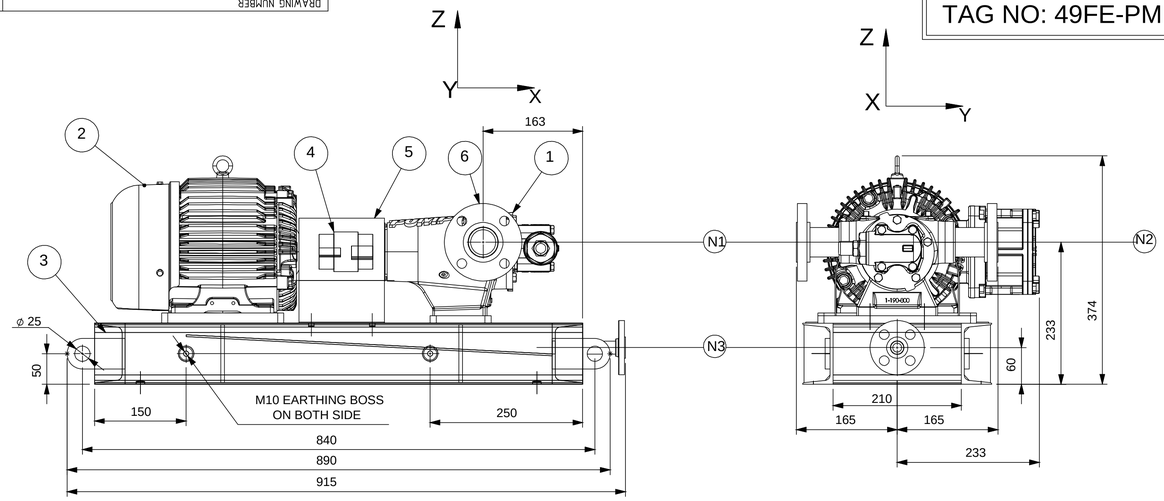
DRAWN	--	--
CHECKED	SRM	10/09/2020
ENGINEER	--	--
APPROVED	KBJ	11/09/2020
EMPLOYER		
SCALE	NTS	A1

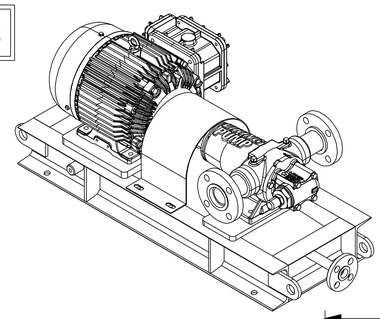
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
FUEL OIL STORAGE
JET FUEL TRANSFER PUMPS

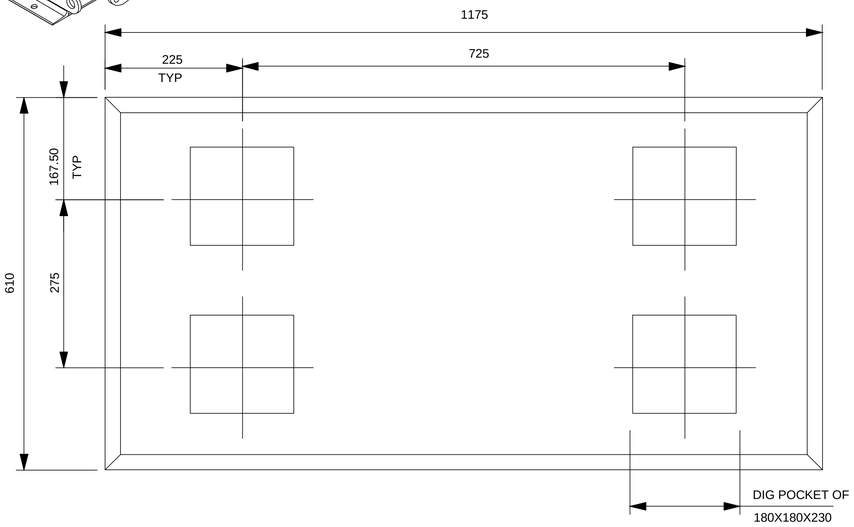
DRAWING NUMBER: MD-216-49FE-EG-ME-GAD-V558-1014 — REV: A02

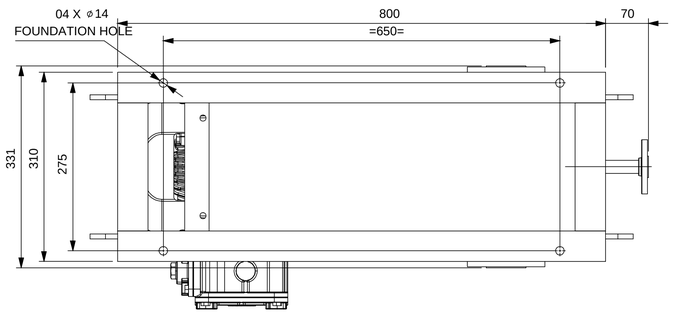
DRAWING NUMBER: MD-216-49FE-EG-ME-GAD-V558-1015 | REV: A01

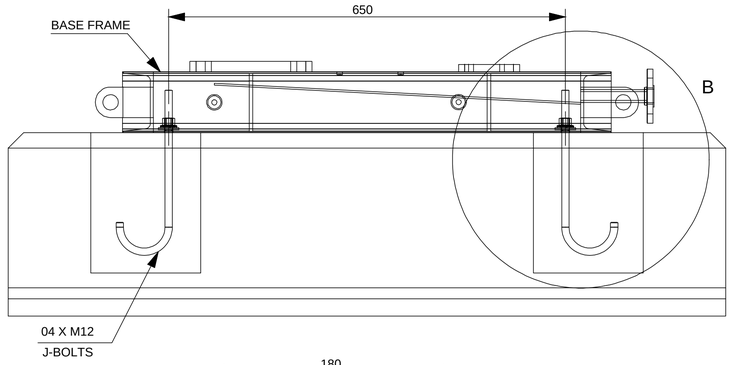
TAG NO: 49FE-PMP-013

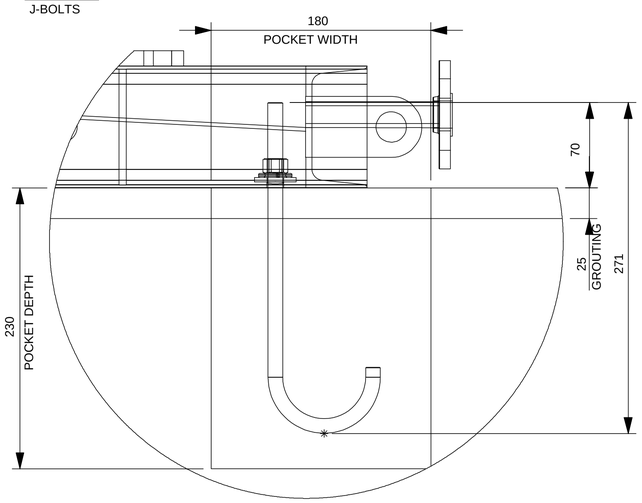












DETAIL B
SCALE 2 : 13

NOTES:

1) ALL DIMENSIONS ARE IN mm.
2) ALL DIMENSIONS TOLERANCE ±5mm.
3) FOUNDATION BOLTS SHALL BE SUPPLIED BY IDEX.

PUMP PERFORMANCE DATA			
PUMP	GD^2 (kg-m^2)	FLOWRATE (LPM)	MAX. DIFF. PR. [bar(g)]
HL 4195	0.0058	113	2.9

MARK	QTY.	SIZE	CLASS	SERVICE
N1	1	1.5" SORF	#150	SUCTION
N2	1	1.5" SORF	#150	DISCHARGE
N3	1	0.5" SORF	#150	DRAIN

ALLOWABLE NOZZLE LOADING (FORCES & MOMENTS)								
NOZZLE NO	NOZZLE	SIZE	FORCE IN NEWTON			MOMENTS IN NEWTON - METER		
			PX	PY	PZ	MX	MY	MZ
N1	SUCTION	1.5"	500	500	500	250	250	35
N2	DISCHARGE	1.5"	500	500	500	250	250	35

ITEM NO.	PART NUMBER	DESCRIPTION	MATERIAL / MAKE	QTY. / SET
1	PUMP	MODEL: HL 4195 TAG NO. : 49FE-PMP-013	SHALL BE AS PER PUMP DATA SHEET. MAKE: VIGEN	1
2	MOTOR	2.2 kW 4 POLE, FOOT MOUNTED TAG: 49FE-PMP-013-MTR	SHALL BE AS PER MOTOR DATA SHEET.	1
3	BASE FRAME	ISMC 100 X 50	GR. E250 BR/ B0	1
4	COUPLING	STD	FENNER	1
5	COUPLING GUARD	PAINTED	CRCA	1

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-49FE-EG-PR-PID-V558-1003	FUEL OIL STORAGE PID FOR DIESEL DISPENSING STATION COMPOUND	A01	11/09/2020	ISSUED FOR REFERENCE	--	SRM	--	KBJ	

ABUNAYYAN-UNIGAZ

DRAWING NUMBER					REVISION
2006-UNI-DWG-056					0
DATE	08.09.2020	DATE	09.09.2020	DATE	10.09.2020
DRAWN	JS	CHECKED	PKS	APPROVED	SR

LARSEN & TOUBRO DRAWING NUMBER	REVISION
O19045_080_U_V_19146_009	0
OUTOTEC DRAWING NUMBER: 900707 __GAU01 __AG50__MTL01 __49025	DOCUMENT ID: OU100694275 — REVISION: 0

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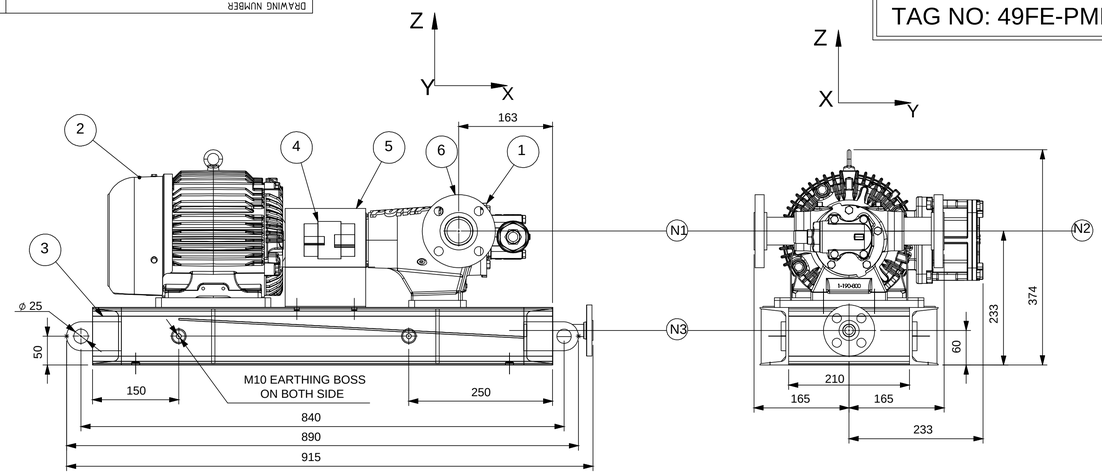
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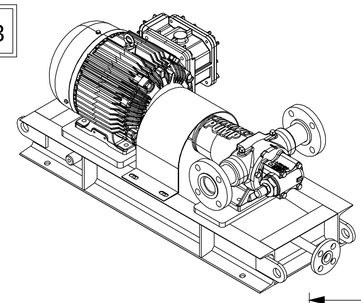
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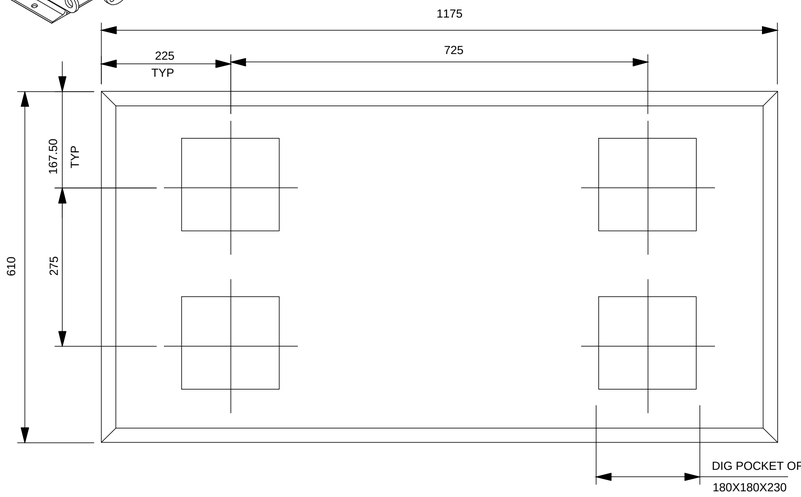
DRAWN	--	--	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SRM	10/09/2020	GENERAL ARRANGEMENT
ENGINEER	--	--	FUEL OIL STORAGE
APPROVED	KBJ	11/09/2020	DRAIN PUMP FOR DIESEL DISPENSING AREA
EMPLOYER			
SCALE	NTS	A1	DRAWING NUMBER: MD-216-49FE-EG-ME-GAD-V558-1015 — REV: A01

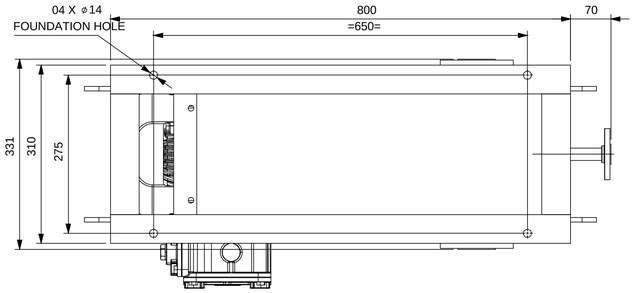
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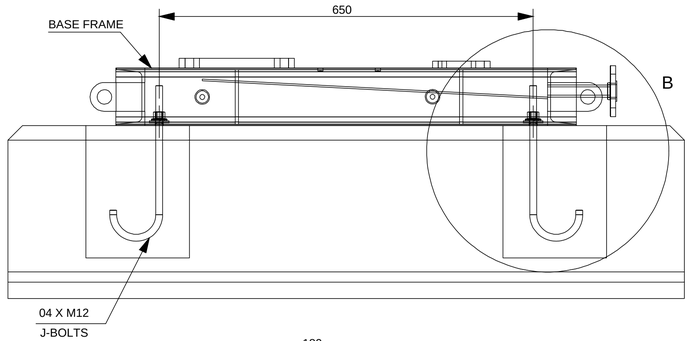
TAG NO: 49FE-PMP-013

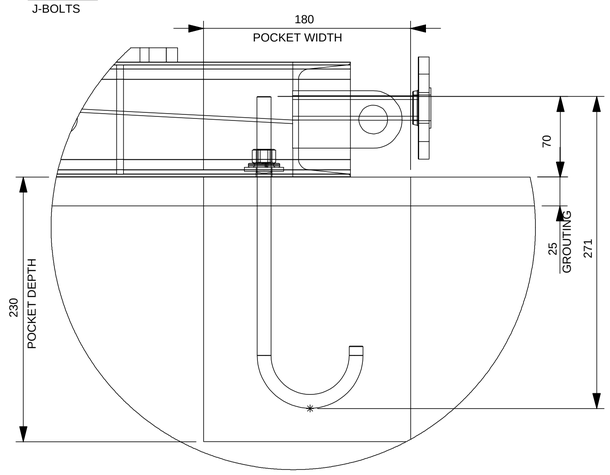












DETAIL B
SCALE 2 : 13

NOTES:

1) ALL DIMENSIONS ARE IN mm.
2) ALL DIMENSIONS TOLERANCE ±5mm.
3) FOUNDATION BOLTS SHALL BE SUPPLIED BY IDEX.

PUMP PERFORMANCE DATA			
PUMP	GD^2 (kg-m^2)	FLOWRATE (LPM)	MAX. DIFF. PR. [bar(g)]
HL 4195	0.0058	113	2.9

MARK	QTY.	SIZE	CLASS	SERVICE
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N2	1	1.5" SORF	#150	DISCHARGE
N3	1	0.5" SORF	#150	DRAIN

ALLOWABLE NOZZLE LOADING (FORCES & MOMENTS)								
NOZZLE NO	NOZZLE	SIZE	FORCE IN NEWTON			MOMENTS IN NEWTON - METER		
			PX	PY	PZ	MX	MY	MZ
N1	SUCTION	1.5"	500	500	500	250	250	35
N2	DISCHARGE	1.5"	500	500	500	250	250	35

ITEM NO.	PART NUMBER	DESCRIPTION C01	MATERIAL / MAKE	QTY. / SET
1	PUMP	MODEL: HL 4195 TAG NO. : 49FE-PMP-013 MAKE: VIGEN	SHALL BE AS PER PUMP DATA SHEET (MA'ADEN DOC. NO. MD-216-49FE-EG-ME-DAT -V558-1001) MAKE : VIGEN	1
2	MOTOR	2.2 kW 4 POLE, FOOT MOUNTED TAG: 49FE-PMP-013-MTR	SHALL BE AS PER MOTOR DATA SHEET (MA'ADEN DOC. NO. MD-216-49FE-EG-EL-DAT- V558-1001)	1
3	BASE FRAME	ISMC 100 X 50	GR. E250 BR/ B0	1
4	COUPLING	STD	FENNER	1
5	COUPLING GUARD	PAINTED	CRCA	1

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		C01	17/10/2020	ISSUED FOR CONSTRUCTION AND REVISED AS PER EMPLOYER COMMENTS DATED 13/09/2020	--	SRM	--	KBJ	
MD-216-49FE-EG-PR-PID-V558-1003	FUEL OIL STORAGE PID FOR DIESEL DISPENSING STATION COMPOUND	A01	11/09/2020	ISSUED FOR REFERENCE	--	SRM	--	KBJ	

ABUNAYYAN-UNIGAZ

DRAWING NUMBER					REVISION
2006-UNI-DWG-056					1
DATE	08.09.2020	DATE	09.09.2020	DATE	10.09.2020
DRAWN	JS	CHECKED	PKS	APPROVED	SR

LARSEN & TOUBRO DRAWING NUMBER: 019045_080_U_V_19146_009 | REVISION: 1
OUTOTEC DRAWING NUMBER: 900707 _GAU01 _AG50_MTL01 _49025 | DOCUMENT ID: OU100694275 | REVISION: 1

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DRAWN	--	--	
CHECKED	SRM	10/09/2020	
ENGINEER	--	--	
APPROVED	KBJ	11/09/2020	
EMPLOYER			
SCALE	NTS	A1	

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
FUEL OIL STORAGE
DRAIN PUMP FOR DIESEL DISPENSING AREA

DRAWING NUMBER: MD-216-49FE-EG-ME-GAD-V558-1015 | REV: C01

DRAWING NUMBER: MD-216-49FE-EG-ME-GAD-V558-1020 | REV: A02

NOZZLE SCHEDULE

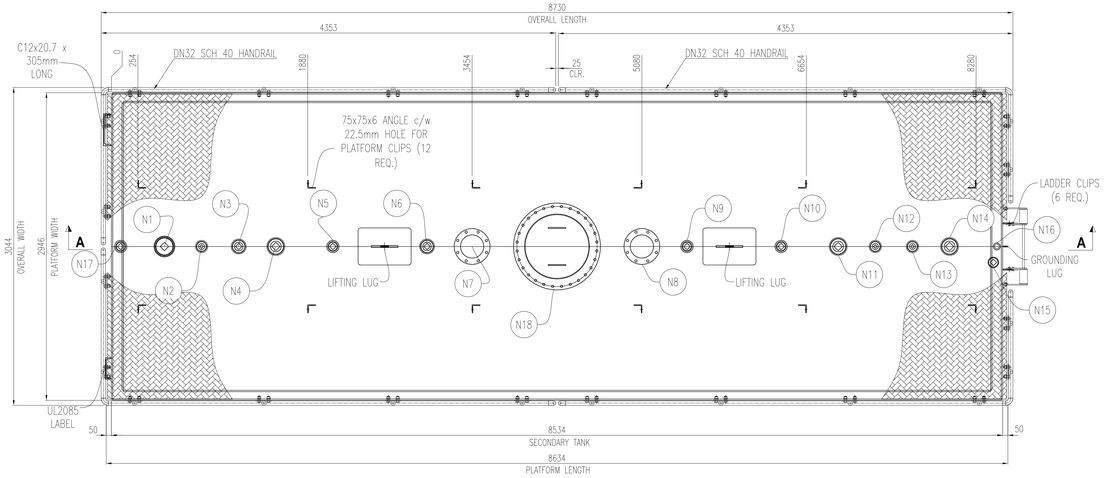
SR. NO.	NOZZLE NO.	SIZE IN INCH	SIZE IN DN (A02)	NOZZLE DESCRIPTION	QTY	SR. NO.	NOZZLE NO.	SIZE IN INCH	SIZE IN DN (A02)	NOZZLE DESCRIPTION	QTY
1	N1	6"ø FPT	DN150 FPT	FUEL FILL (c/w DN80 OVERFILL PREVENTION VALVE)	1	10	N10	3"ø FPT	DN80 FPT	INSULATION PORT	1
2	N2	2"ø FPT	DN50 FPT	SPARE	1	11	N11	4"ø FPT	DN100 FPT	TANK OUTLET (c/w DN40 DROP TUBE & FOOT VALVE)	1
3	N3	3"ø FPT	DN80 FPT	FLAME ARRESTER	1	12	N12	2"ø FPT	DN50 FPT	LEVEL SWITCH	1
4	N4	4"ø FPT	DN100 FPT	SUBMERSIBLE TURBINE PUMP (FRANKLIN, USA)	1	13	N13	2"ø FPT	DN50 FPT	TANK GAUGING SYSTEM (WITH TRANSMITTER)	1
5	N5	3"ø FPT	DN80 FPT	INSULATION PORT	1	14	N14	4"ø FPT	DN100 FPT	SPARE	1
6	N6	3"ø FPT	DN80 FPT	PRIMARY TANK NORMAL VENT	1	15	N15	3"ø H. CPLG	DN80 FPT	INSULATION PORT	1
7	N7	8"ø FLG.	DN200 FPT	SECONDARY TANK EMERGENCY VENT	1	16	N16	2"ø MPT	DN50 FPT	INTERSTITIAL SPACE LEAK DETECTOR PIPE	1
8	N8	8"ø FLG.	DN200 FPT	PRIMARY TANK EMERGENCY VENT	1	17	N15	3"ø FPT	DN80 FPT	INSULATION PORT	1
9	N9	3"ø FPT	DN80 FPT	SECONDARY TANK NORMAL VENT	1	18	N16	24"ø	DN600 FPT	MANWAY	1

CONSTRUCTION DATA FOR 50 M^3 DIESEL TANK (TAG NUMBER: 49FE-TNK-006)

DESIGN & FABRICATION STANDARD		UL2085 (A02)	
MAX. DESIGN TEMP IN °C:	93.3 °C	PRIMARY TANK ID IN mm:	2,769mm
WORKING PRESSURE IN kPa: (A02)	ATMOSPHERIC (99.9kPa)	SECONDARY TANK ID IN mm:	2,934mm
DESIGN PRESSURE IN kPa:	N/A (A02)	TOTAL VOLUME IN LITRES:	50,000 LITRES
PNEUMATIC TEST PRESSURE IN kPa:	20-34 kPa	PRIMARY HEAD THICKNESS IN mm:	7.93mm
PNEUMATIC TEST TEMPERATURE IN °C:	AMBIENT	SECONDARY HEAD THICKNESS IN mm:	7.93mm
PRIMARY SHELL THICKNESS IN mm:	6.35mm	PRIMARY SHELL LENGTH IN mm:	8,331mm
SECONDARY SHELL THICKNESS IN mm:	6.35mm	SECONDARY SHELL LENGTH IN mm:	8,534mm

LIST OF MATERIAL SPECIFICATION

DESCRIPTION	MATERIAL SPECIFICATION
SHELL	MIN. ASTM A36
FLAT HEAD	MIN. ASTM A36
NOZZLES	ASTM A105 & ASTM A181-60
NOZZLE NECK/PIPE	ASTM A53 GR. B
MANHOLE	MIN. ASTM A36



TOP VIEW

FIREGUARD FITTINGS

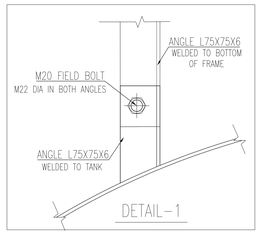
(3) DN50 FPT c/w PLUG & DN80 SCH. 40 PIPE X 178mm
(2) DN80 FPT c/w PLUG & DN100 SCH. 40 PIPE X 178mm
(3) DN100 FPT c/w PLUG & DN125 SCH. 40 PIPE X 178mm
(1) DN150 H.CPG c/w PLUG & 150 SCH. 40 PIPE X 165mm

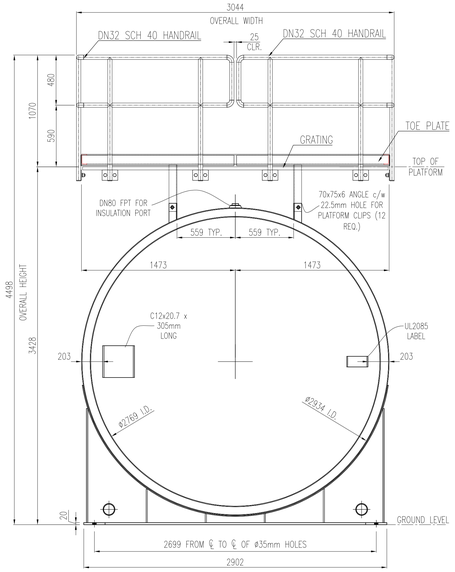
PAINT NOTES:

SANDBLAST EXTERIOR & APPLY ONE COAT OF EPOXY & ONE COAT OF "RAL 9010" POLYURETHANE

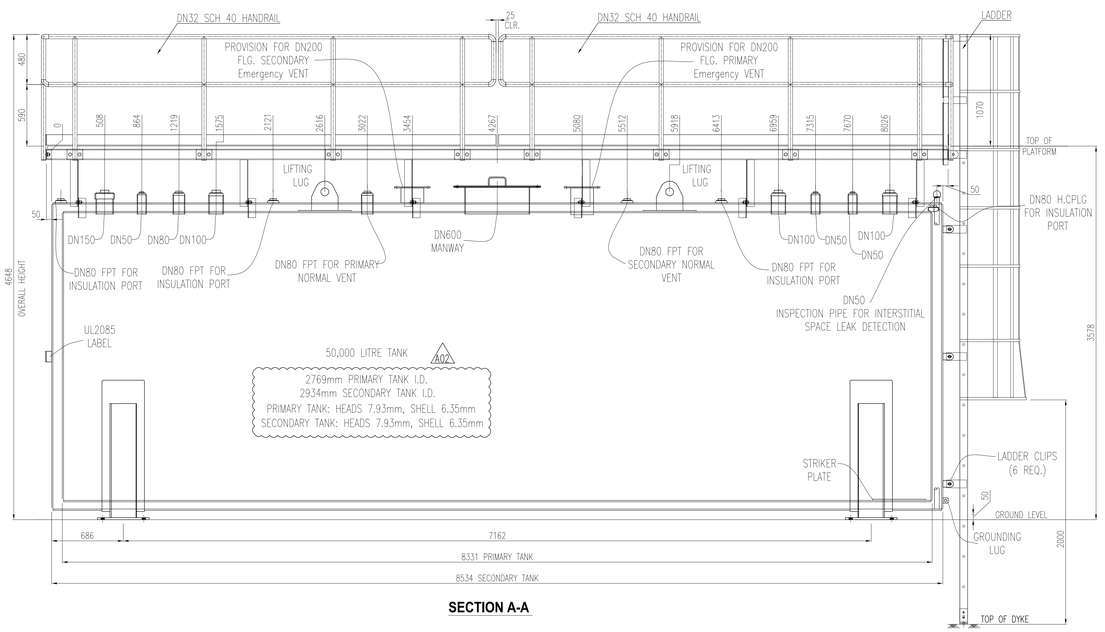
SHIPPED LOOSE:

- (2) DN80 x 610mm LONG ALUM. VENT PIPE
- (2) DN80 FIG. 354 UPDRAFT VENT
- (2) DN200 FIG. 244 FLG. EMERGENCY VENT (A02)
- (1) DN100x DN40 DOUBLE TAPPED BUSHING c/w DN40 DROP TUBE & DN40 SINGLE POPPET FOOT VALVE @ 100mm FROM TANK BOTTOM



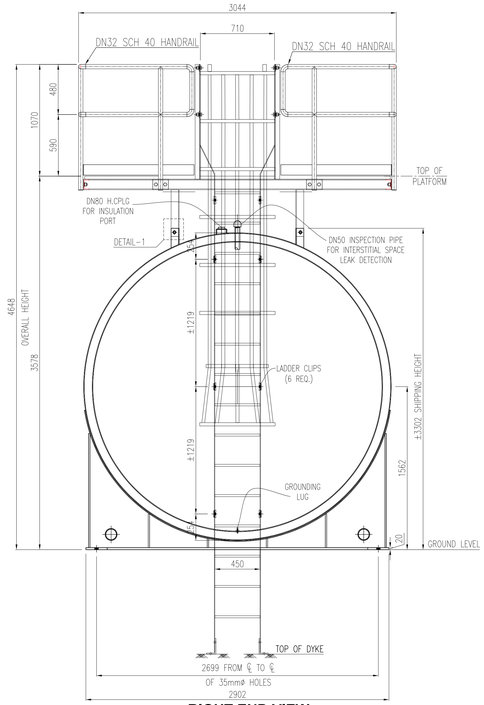


LEFT END VIEW



SECTION A-A

TANK TAG NUMBER: 49FE-TNK-006



RIGHT END VIEW

EMPTY SHIPPING WEIGHT: 15,850 kg
WEIGHT WITH CONTENTS: 65,975 kg

NOTE:

1. ALL DIMENSIONS ARE IN MILLIMETER UNLESS OTHERWISE SPECIFIED.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A02	07/11/2020	REVISED AS PER EMPLOYER COMMENTS DATED 24/09/2020 & AS MARKED	---	SRM	---	KBJ	
MD-216-49FE-EG-PR-PID-V558-1003	FUEL OIL STORAGE PID FOR DIESEL DISPENSING STATION COMPOUND	A01	01/09/2020	ISSUED FOR REFERENCE	---	SRM	---	KBJ	

ABUNAYYAN-UNIGAZ

DRAWING NUMBER	REVISION
2006-UNI-DWG-021	1

DATE		DATE		DATE	
27.08.2020	29.08.2020			30.08.2020	
DRAWN	JD	CHECKED	PKS	APPROVED	RS

LARSEN & TOUBRO DRAWING NUMBER: 019045_080_U_V_19148_004 — REVISION: 1
OUTOTEC DRAWING NUMBER: 900707__GAU01__AG50_MTL01__49030 — DOCUMENT ID: OU100694280 — REVISION: 1

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DRAWN	---	---	
CHECKED	SRM	01/09/2020	
ENGINEER	---	---	
APPROVED	KBJ	01/09/2020	
EMPLOYER			
SCALE	NTS	A0	

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
FUEL OIL STORAGE
TANK (50 M^3) FOR DIESEL DISPENSING SYSTEM

DRAWING NUMBER: MD-216-49FE-EG-ME-GAD-V558-1020 — REV: A02

NOZZLE SCHEDULE (A02)					
SR. NO.	NOZZLE NO.	SIZE IN DN	SIZE IN INCHES	NOZZLE DESCRIPTION	QTY
1	N1	DN150 FPT	6"ø FPT	FUEL FILL (c/w DN80 OVERFILL PREVENTION VALVE)	1
2	N2	DN50 FPT	2"ø FPT	CABLE TO FILL POINT CABINET	1
3	N3	DN100 FPT	4"ø FPT	SPARE	1
4	N4	DN 80 FPT	3"ø FPT	FLAME ARRESTER	1
5	N5	DN80 FPT	3"ø FPT	INSULATION PORT	1
6	N6	DN80 FPT	3"ø FPT	PRIMARY TANK NORMAL VENT	1
7	N7	DN200 FLG.	8"ø FLG.	SECONDARY TANK EMERGENCY VENT	1
8	N8	DN200 FLG.	8"ø FLG.	PRIMARY TANK EMERGENCY VENT	1
9	N9	DN80 FPT	3"ø FPT	SECONDARY TANK NORMAL VENT	1
10	N10	DN80 FPT	3"ø FPT	INSULATION PORT	1
11	N11	DN50 FPT	2"ø FPT	TANK GAUGING SYSTEM (WITH TRANSMITTER)	1
12	N12	DN50 FPT	2"ø FPT	LEVEL SWITCH	1
13	N13	DN 100 FPT	4"ø FPT	TANK OUTLET (c/w DN40 DROP TUBE & FOOT VALVE)	1
14	N14	DN100 FPT	4"ø FPT	SUBMERSIBLE TURBINE PUMP (FRANKLIN, USA)	1
15	N15	DN80 H.CPLG	3"ø H. CPLG	INSULATION PORT	1
16	N16	DN50 MPT	2"ø MPT	INTERSTITIAL SPACE LEAK DETECTOR PIPE	1
17	N17	DN80 FPT	3"ø FPT	INSULATION PORT	1
18	N18	DN600	24"ø	MANWAY c/w INTERNAL LADDER	1

CONSTRUCTION DATA FOR 35 M^3 PETROL TANK (TAG NUMBER: 49FE-TNK-003)			
DESIGN & FABRICATION STANDARD:		UL2085 (A02)	
MAX. DESIGN TEMP IN °C:	93.3 °C	PRIMARY TANK ID IN mm:	2,515mm
WORKING PRESSURE IN kPa: (A02)	ATMOSPHERIC (A01) (A02)	SECONDARY TANK ID IN mm:	2,680mm
DESIGN PRESSURE IN kPa:	N/A	TOTAL VOLUME IN LITRES:	35,000 LITRES
PNEUMATIC TEST PRESSURE IN kPa: (A02)	20.7 (A02)	PRIMARY HEAD THICKNESS IN mm:	7.93mm
PNEUMATIC TEST TEMPERATURE IN °C:	AMBIENT	SECONDARY HEAD THICKNESS IN mm:	7.93mm
PRIMARY SHELL THICKNESS IN mm:	6.35mm	PRIMARY SHELL LENGTH IN mm:	7,112mm
SECONDARY SHELL THICKNESS IN mm:	6.35mm	SECONDARY SHELL LENGTH IN mm:	7,315mm

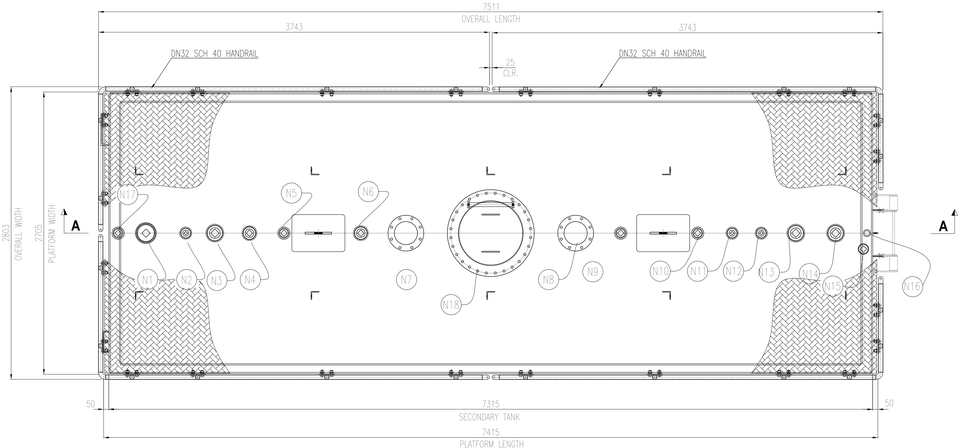
LIST OF MATERIAL SPECIFICATION	
DESCRIPTION	MATERIAL SPECIFICATION
SHELL	MIN. ASTM A36
FLAT HEAD	MIN. ASTM A36
NOZZLES	ASTM A105 & ASTM A181-60
NOZZLE NECK/PIPE	ASTM A53 GR. B
MANHOLE	MIN. ASTM A36

EMPTY SHIPPING WEIGHT: 12,135 kg
WEIGHT WITH CONTENTS: 48,195 kg

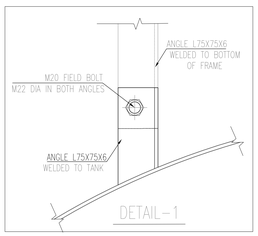
PAINT NOTES:
SANDBLAST EXTERIOR & APPLY ONE COAT OF EPOXY & ONE COAT OF "RAL 9010" POLYURETHANE

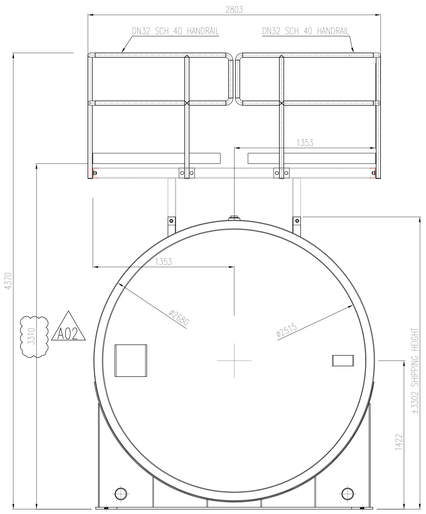
FIREGUARD FITTINGS
- (4) DN50 FPT c/w PLUG & DN80 SCH. 40 PIPE X 178mm
- (2) DN80 FPT c/w PLUG & DN100 SCH. 40 PIPE X 178mm
- (2) DN100 FPT c/w PLUG & DN125 SCH. 40 PIPE X 178mm
- (1) DN150 H.CPG c/w PLUG & DN150 SCH. 40 PIPE X 165mm

SHIPPED LOOSE:
- (2) DN80 x 1220mm LONG ALUM. VENT PIPE
- (2) DN80 FIG. 354 UPDRAFT VENT
- (2) DN200 FIG. 244 FLG. EMERGENCY VENT (A02)
- (1) DN100x DN40 DOUBLE TAPPED BUSHING c/w DN40 DROP TUBE & DN40 SINGLE POPPET FOOT VALVE @ DN100 FROM TANK BOTTOM

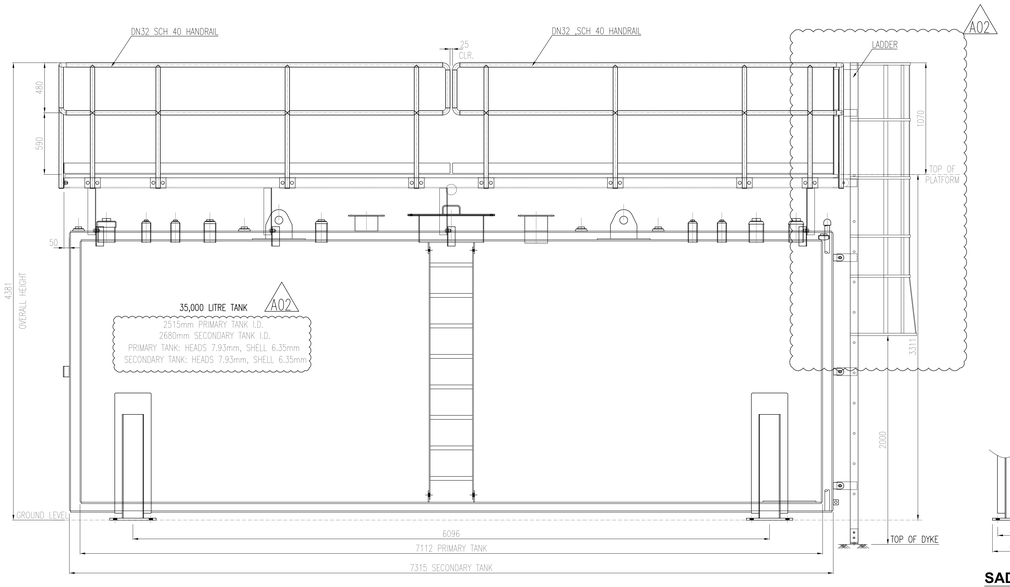


TOP VIEW





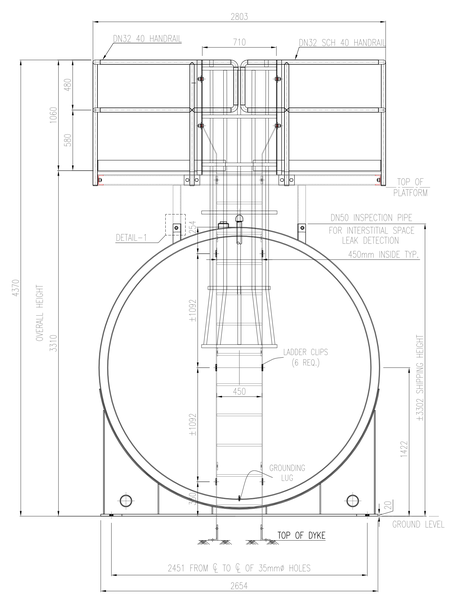
LEFT END VIEW



SECTION A-A



SADDLE SIDE VIEW



RIGHT END VIEW

TANK TAG NUMBER: 49FE-TNK-003

NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER UNLESS OTHERWISE SPECIFIED.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-49FE-PR-PID-V558-1009	FUEL OIL STORAGE PID FOR PETROL UNLOADING, STORAGE & DISTRIBUTION	A02	07/11/2020	REVISED AS PER EMPLOYER COMMENTS DATED 02/10/2020 & AS MARKED	---	SRM	---	KBJ	
		A01	05/09/2020	ISSUED FOR REFERENCE	---	SRM	---	KBJ	

ABUNAYYAN-UNIGAZ

DRAWING NUMBER					REVISION
2006-UNI-DWG-022					1
DATE	27.08.2020	DATE	29.08.2020	DATE	30.08.2020
DRAWN	JD	CHECKED	PKS	APPROVED	RS

LARSEN & TOUBRO DRAWING NUMBER		REVISION
019045_080_U_V_19148_005		1
OUTOTEC DRAWING NUMBER	DOCUMENT ID	REVISION
900707 __ GAU01 __ AG50__ MTL01 __49031	OU100694281	1

DRAWN	---	---
CHECKED	SRM	05/09/2020
ENGINEER	---	---
APPROVED	KBJ	05/09/2020
EMPLOYER		
SCALE	NTS	A0

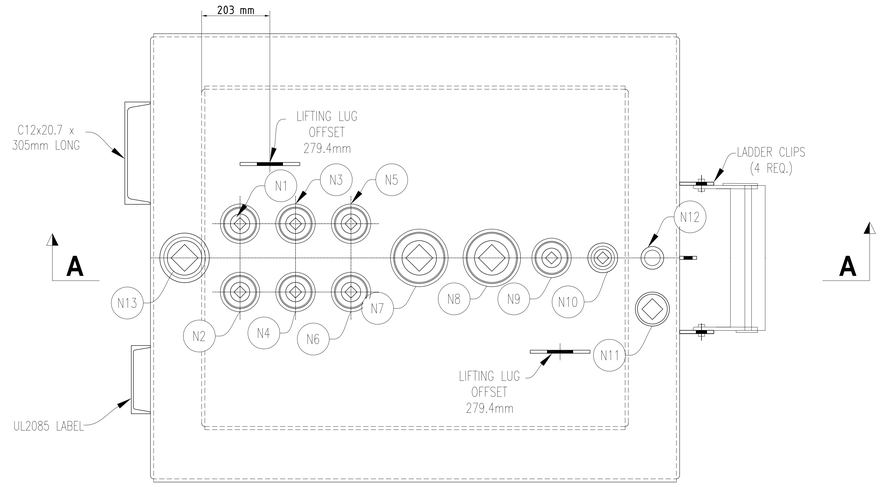
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
FUEL OIL STORAGE
TANK (35 M^3) FOR PETROL DISPENSING SYSTEM

DRAWING NUMBER: MD-216-49FE-EG-ME-GAD-V558-1021 REV: A02

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TOP VIEW

NOZZLE SCHEDULE (A02)					
SR. NO.	NOZZLE NO.	SIZE in DN	SIZE in inhces	NOZZLE DESCRIPTION	QTY
1	N1	DN50 FPT	2"ø FPT	FUEL FILL	1
2	N2	DN50 FPT	2"ø FPT	CABLE TO FILL POINT CABINET	1
3	N3	DN50 FPT	2"ø FPT	SPARE	1
4	N4	DN50 FPT	2"ø FPT	FLAME ARRESTER	1
5	N5	DN50 FPT	2"ø FPT	TANK GAUGING SYSTEM (WITH TRANSMITTER)	1
6	N6	DN50 FPT	2"ø FPT	LEVEL SWITCH	1
7	N7	DN100 FPT	4"ø FPT	TANK OUTLET (c/w DN50 DROP TUBE & FOOT VALVE)	1
8	N8	DN100 FPT	4"ø FPT	PRIMARY TANK EMERGENCY VENT	1
9	N9	DN50 FPT	2"ø FPT	PRIMARY TANK NORMAL VENT	1
10	N10	DN50 FPT	2"ø FPT	SECONDARY TANK NORMAL VENT	1
11	N11	DN80 H. CPLG	3"ø H. CPLG	INSULATION PORT	1
12	N12	DN50 MPT	2"ø MPT	INTERSTITIAL SPACE LEAK DETECTOR PIPE	1
13	N13	DN100 FPT	4"ø FPT	INSULATION PORT & SECONDARY TANK EMERGENCY VENT	1

CONSTRUCTION DATA CONSTRUCTION DATA FOR 1 M^3 JET FUEL TANK (TAG NUMBER: 49FE-TNK-009)			
DESIGN & FABRICATION STANDARD	UL2085 (A02)		
MAX. DESIGN TEMP IN °C:	93.3 °C	PRIMARY TANK ID IN mm:	1,016mm
WORKING PRESSURE IN kPa: (A02)	ATMOSPHERIC (99.9kPa)	SECONDARY TANK ID IN mm:	1,328mm
DESIGN PRESSURE IN kPa:	N/A (A02)	TOTAL VOLUME IN LITRES:	1,000 LITRES
PNEUMATIC TEST PRESSURE IN kPa:	20–34 kPa	PRIMARY HEAD THICKNESS IN mm:	3.42mm
PNEUMATIC TEST TEMPERATURE IN °C:	AMBIENT	SECONDARY HEAD THICKNESS IN mm:	3.42mm
PRIMARY SHELL THICKNESS IN mm:	3.42mm	PRIMARY SHELL LENGTH IN mm:	1,270mm
SECONDARY SHELL THICKNESS IN mm:	3.42mm	SECONDARY SHELL LENGTH IN mm:	1,575mm

LIST OF MATERIAL SPECIFICATION	
DESCRIPTION	MATERIAL SPECIFICATION
SHELL	MIN. A36/44W
FLAT HEAD	MIN. A36/44W
NOZZLES	ASTM A105 & ASTM A181-60
NOZZLE NECK/PIPE	ASTM A53 GR. B
MANHOLE	MIN. A36/44W

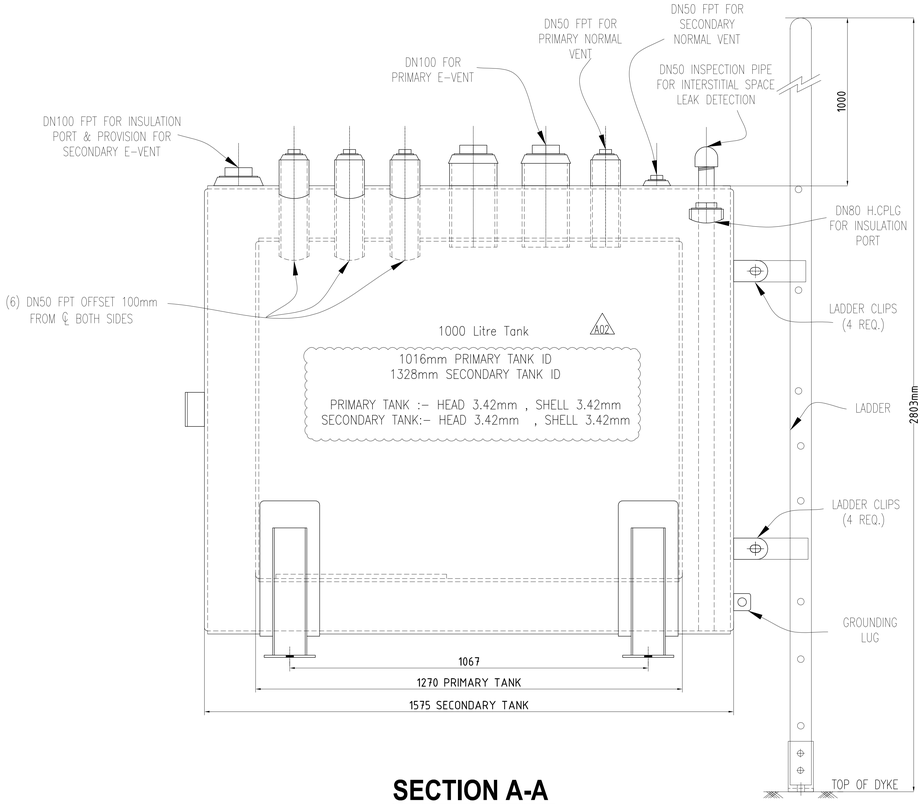
EMPTY SHIPPING WEIGHT: 1,385 kg
WEIGHT WITH CONTENTS: 2,395 kg

SHIPPED LOOSE:
- (2) DN50 x 2438 mm LONG ALUM. VENT PIPE
- (2) DN50 FIG. 354 UPDRAFT VENT
- (2) DN100 FIG. 244 THR. EMERGENCY-VENT (A02)
- (2) DN50x25 FIG. 184 DOUBLE TAPPED BUSHING
- (1) DN100x50 DOUBLE TAPPED BUSHING c/w DN50 DROP TUBE & DN50 SINGLE POPPET FOOT VALVE @ DN100 FROM TANK BOTTOM

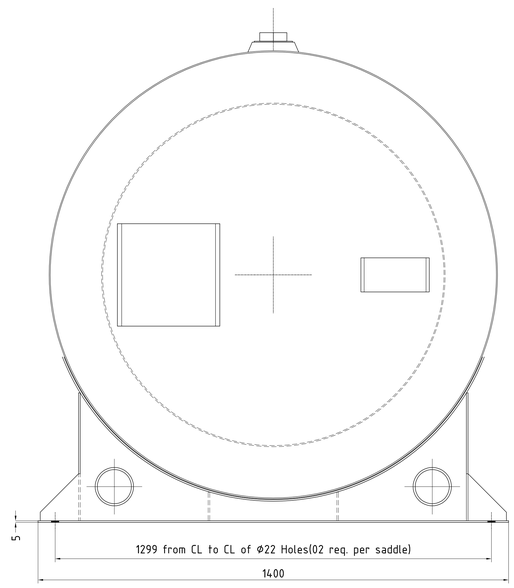
PAINT NOTES:
SANDBLAST EXTERIOR & APPLY ONE COAT OF EPOXY & ONE COAT OF "RAL 9010" POLYURETHANE

OFFSET FIREGUARD FITTINGS:
(6) DN50 FPT c/w PLUG & DN80 SCH. 40 PIPE x 279mm

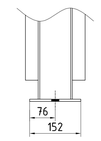
CENTERLINE FIREGUARD FITTINGS
(1) DN50 FPT c/w PLUG & DN80 SCH. 40 PIPE X 254mm
(2) DN100 FPT c/w PLUG & DN125 SCH. 40 PIPE X 254mm



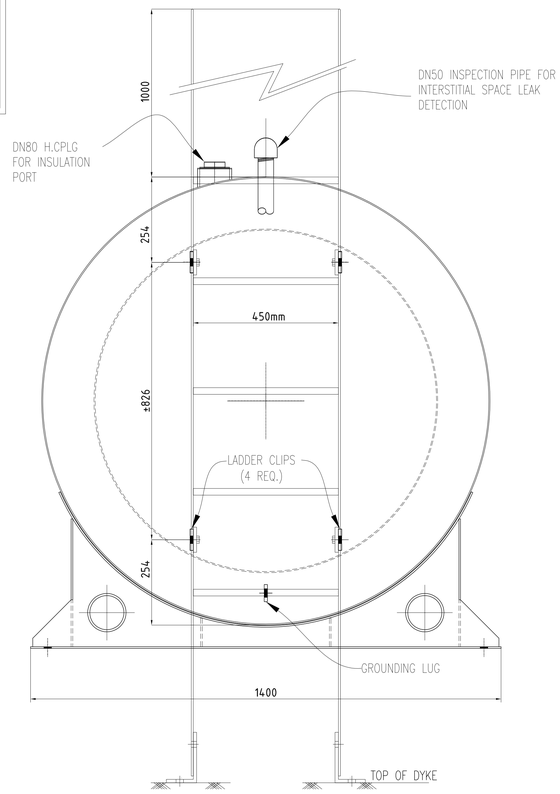
SECTION A-A



LEFT END VIEW



SADDLE SIDE VIEW



RIGHT END VIEW

TANK TAG NUMBER: 49FE-TNK-009

NOTE:

1. ALL DIMENSIONS ARE IN MILLIMETER UNLESS OTHERWISE SPECIFIED.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A02	31/10/2020	REVISED AS PER EMPLOYER COMMENTS DATED 02/10/2020 & AS MARKED	---	SRM	---	KBJ	
MD-216-49FE-EG-PR-PID-V558-1005	FUEL OIL STORAGE PID FOR JET FUEL UNLOADING, STORAGE & DISTRIBUTION	A01	05/09/2020	ISSUED FOR REFERENCE	---	SRM	---	KBJ	

ABUNAYYAN-UNIGAZ

DRAWING NUMBER					REVISION
2006-UNI-DWG-023					1
DATE	01.09.2020	DATE	02.09.2020	DATE	03.09.2020
DRAWN	JD	CHECKED	PKS	APPROVED	RS

LARSEN & TOUBRO DRAWING NUMBER			REVISION
019045_080_U_V_19148_006			1
OUTOTEC DRAWING NUMBER		DOCUMENT ID	REVISION
900707__GAU01__AG50__MTL01__49032		OU100694282	1
DRAWN	---	---	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SRM	05/09/2020	GENERAL ARRANGEMENT
ENGINEER	---	---	FUEL OIL STORAGE
APPROVED	KBJ	05/09/2020	TANK (1 M^3) FOR JET FUEL DISTRIBUTION SYSTEM
EMPLOYER			
SCALE	NTS	A0	DRAWING NUMBER: MD-216-49FE-EG-ME-GAD-V558-1022 REV: A02

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ATTACHMENT 3 THERMAL EXPANSION COEFFICIENT

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Total linear expansion between 70°F and indicated temperature (inches/100 ft)

Temp. (°F)	Shell Material								Temp. (°F)
	Carbon Steel Carbon - Moly Low-Chrome (through 3 Cr Mo)	5 Cr Mo through 9 Cr Mo	Austenitic Stainless Steels 18 Cr 8 Ni	12 Cr 17 Cr 27 Cr	12 Cr 20 Ni	Monel 67 Ni 30 Cu	3-1/2 Nickel	Ni-Fe-Cr	
70	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	70
100	0.23	0.22	0.34	0.20	0.32	0.28	0.23	0.28	100
125	0.42	0.40	0.62	0.36	0.58	0.52	0.42	0.52	125
150	0.61	0.58	0.90	0.53	0.84	0.75	0.61	0.76	150
175	0.80	0.76	1.18	0.69	1.10	0.99	0.81	0.99	175
200	0.99	0.94	1.46	0.86	1.37	1.22	1.01	1.23	200
225	1.21	1.13	1.75	1.03	1.64	1.46	1.21	1.49	225
250	1.40	1.33	2.03	1.21	1.91	1.71	1.42	1.76	250
275	1.61	1.52	2.32	1.38	2.18	1.96	1.63	2.03	275
300	1.82	1.71	2.61	1.56	2.45	2.21	1.84	2.30	300
325	2.04	1.90	2.90	1.74	2.72	2.44	2.05	2.59	325
350	2.26	2.10	3.20	1.93	2.99	2.68	2.26	2.88	350
375	2.48	2.30	3.50	2.11	3.26	2.91	2.47	3.18	375
400	2.70	2.50	3.80	2.30	3.53	3.25	2.69	3.48	400
425	2.93	2.72	4.10	2.50	3.80	3.52	2.91	3.76	425
450	3.16	2.93	4.41	2.69	4.07	3.79	3.13	4.04	450
475	3.39	3.14	4.71	2.89	4.34	4.06	3.35	4.31	475
500	3.62	3.35	5.01	3.08	4.61	4.33	3.58	4.59	500
525	3.86	3.58	5.31	3.28	4.88	4.61	3.81	4.87	525
550	4.11	3.80	5.62	3.49	5.15	4.90	4.04	5.16	550
575	4.35	4.02	5.93	3.69	5.42	5.18	4.27	5.44	575
600	4.60	4.24	6.24	3.90	5.69	5.46	4.50	5.72	600
625	4.86	4.47	6.55	4.10	5.96	5.75	4.74	6.01	625
650	5.11	4.69	6.87	4.31	6.23	6.05	4.98	6.30	650
675	5.37	4.92	7.18	4.52	6.50	6.34	5.22	6.58	675
700	5.63	5.14	7.50	4.73	6.77	6.64	5.46	6.88	700
725	5.90	5.38	7.82	4.94	7.04	6.94	5.70	7.17	725
750	6.16	5.62	8.15	5.16	7.31	7.25	5.94	7.47	750
775	6.43	5.86	8.47	5.38	7.58	7.55	6.18	7.76	775
800	6.70	6.10	8.80	5.60	7.85	7.85	6.43	8.06	800
825	6.97	6.34	9.13	5.82	8.15	8.16	6.68	8.35	825
850	7.25	6.59	9.46	6.05	8.45	8.48	6.93	8.66	850
875	7.53	6.83	9.79	6.27	8.75	8.80	7.18	8.95	875
900	7.81	7.07	10.12	6.49	9.05	9.12	7.43	9.26	900
925	8.08	7.31	10.46	6.71	9.35	9.44	7.68	9.56	925
950	8.35	7.56	10.80	6.94	9.65	9.77	7.93	9.87	950
975	8.62	7.81	11.14	7.17	9.95	10.09	8.17	10.18	975